Self-Referential Dissipative Systems: A Materials Test Across Ten Substrates

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In memory of my grandfather Franz Scheßl (1929–2024)

Abstract

The same sigmoid functional form recurs across ten physically heterogeneous substrates, read off from a single cumulative measurement quantity along an ordered load history. The finding is grounded in the tensile test: on fiber-parallel loaded spruce specimens (***Picea abies***, DIN 52188:1979-05, tension per DIN EN 408:2012-10), the nonlinearity residual $Q(x) = |\sigma - E\varepsilon| / \max$ of the stress-strain curve normalized to $[0, 1]$, taken over $x = \sigma/\sigma_{\max}$, fits a logistic sigmoid whose inflection location $\hat{a}$ reads the load-bearing capacity. Across $n = 230$ specimens this location correlates with fracture stress at Pearson $r = -0.83$ ($p < 10^{-30}$); the mean inflection location $\tilde{a} = 0.567$ carries the fiber anisotropy, an early inflection signals lower fracture stress. The frame is materials testing applied isomorphically to ordered load histories: the specimen bears a load, accumulates damage, passes a yield point, and becomes irreversible. Every conversation, market trajectory, or autocratization episode is such a specimen, not a bag of features. The burden of proof is thus reversed: a falsification would have to exhibit a finding that violates the mechanics (§8).

The class of self-referential dissipative systems (SRDS) fixes five axioms A0–A4 (self-reference, dissipation, accumulation, transition, irreversibility); only A0 and A1 are primitive, A2–A4 follow as theorems under four regularity conditions L1–L4 (load-bearing L2, the single-peaked state-dependent damage rate, and L4, the return-and-elasticity protocol). Three core results carry it: T1 signs the conditional phase-space volume rate $\mathrm{Tr}\, A$ through the sign of the windowed trend of mutual information; T2 secures, under contraction, a unique non-equilibrium steady state; T3 (Goldilocks principle) places optimal function near criticality; the inflection is the order parameter. A computer-assisted no-go theorem excludes a universal numeric inflection constant ($\dot{Q}(1/2) \in [1.8412, 1.8963]$, Arb certificate, Lean-4 public): what is universal is the functional form, not the numeric value; the inflection location is substrate-specifically measured, not a universal constant. Under substrate-specific normalization the class family carries across timber, multiaxial fatigue ($n = 914$, 136 materials; $\hat{a}$ reads the load-path non-proportionality), financial crashes (crash $R^2 = 0.973$ vs. calm 0.483), lignin (boundary case without an inner $\hat{a}$; $\rho_S = -0.78$), V-Dem autocratization episodes ($n = 117$, median $R^2 = 0.983$), earthquakes ($n = 284$, $\rho_S = +0.673$ Bonferroni), battery degradation ($n = 26$, KS $D = 0.733$), colorectal carcinomas ($n = 579$), and music ($n = 2{,}840$, $\tau = +1.000$). For conversation the load history (lag-1 $\bar{\rho}_1 = 0.928$) is signal, not a nuisance term, and five axes coupled via $\sigma = E\varepsilon$ are read anti-sparse: across $n = 202$ specimens $\hat{a}$ correlates univariately with manipulation ($\rho = -0.359$, cluster-robust), the chat-level ridge reaches $\rho = 0.700 \pm 0.076$, replicated across twelve corpora ($\approx$ 35,300 dialogues from 14 sources). Synthetic null specimens separate sharply (KS $p < 10^{-3}$ on all five substrates tested), a pre-registered null control (ProsocialDialog) and two falsified positive predictions (Sotopia, Google COVID Mobility) map the scope conditions, six of the ten substrates remain Bonferroni-compliant under the safety factor γ_M. Lean-4 proofs, metric code, and the gold corpus are open and reproducible; the operational calibration thresholds remain withheld (responsible disclosure): no finding that carries the cross-substrate thesis depends on them. The structural load-bearing capacity is threshold-independent.

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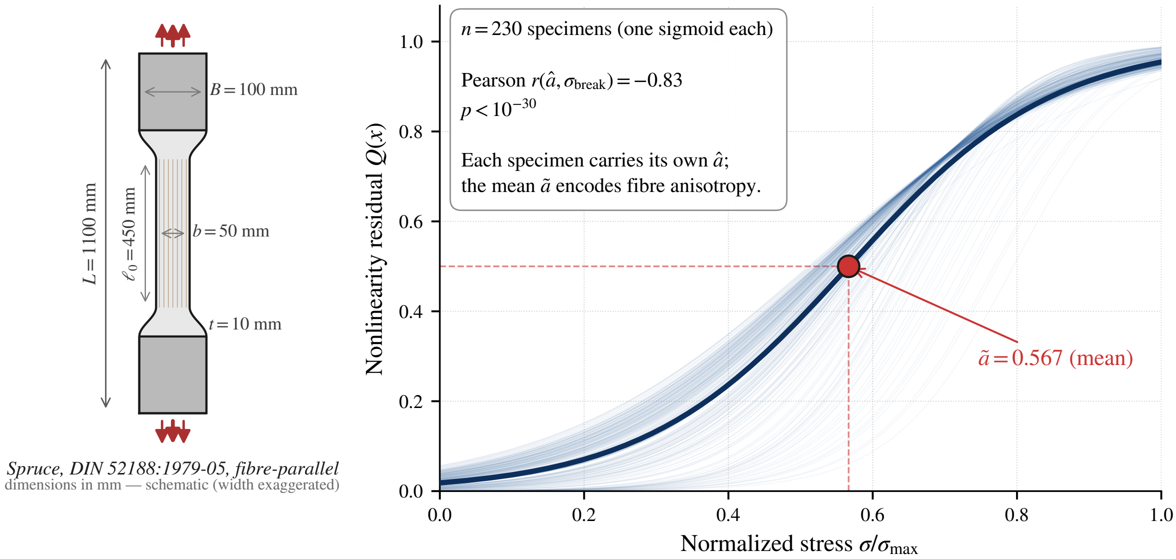
Keywords: large language models, conversation analysis, AI safety, cross-domain validation, self-referential dissipative systems, sigmoid inflection, falsifiability.

Contents

The physical anchor: tensile test per DIN EN 408



Whoever yields early breaks at lower load. The inflection point $\hat{a}$ determines fracture stress.

Figure 1: Bone-shaped timber specimen (spruce, DIN 52188:1979-05, fiber-parallel) under tension per DIN EN 408:2012-10. Right: nonlinearity residual $Q(x)$ with sigmoid fit across $n = 230$ specimens; $\hat{a}$ correlates with fracture stress at $r = -0.83$, $p < 10^{-30}$, mean inflection $\tilde{a} = 0.567$.

1 Question and Anchor

1.1 The Physical Experiment

A spruce timber specimen, bone-shaped per DIN 52188:1979-05, with a cross-section of 50×10 mm in the test region, was tested fiber-parallel under tension per DIN EN 408:2012-10 (Figure 1). The stress-strain curve follows the familiar sequence of linear-elastic regime, yield point, plastic deformation, necking, and fracture; a material holds because it responds to stress: by yielding and redistributing load. Across $n = 230$ specimens we fit the nonlinearity residual $Q = |\sigma - E\varepsilon|/\max$ with a logistic sigmoid $Q(x)$ over the normalized stress $x = \sigma/\sigma_{\max}$ and extract its inflection point $\hat{a}$ ($\tilde{a} = 0.567$); $\hat{a}$ correlates with fracture stress at Pearson $r = -0.83$, $p < 10^{-30}$. An early inflection corresponds to a lower fracture stress. This relationship is established in timber engineering. The question of this work is why the same functional form recurs in nine further substrates: metals under cyclic load, markets, lignin pulping, democratic indices, earthquake sequences, battery trajectories, tumor mutation loads, LLM conversations, musical note-adjacency networks.

1.2 The Paradigm: Materials Testing, Applied Isomorphically

The wood tensile test is not an illustration but the construct anchor. SRDS applies the rules of continuum mechanics and statics isomorphically to sequences under load: to a conversation, a market trajectory, an autocratization episode as much as to a test specimen. The claim is not that language *behaves like* a material, but that the same structural laws (stress–strain coupling, damage accumulation, support reaction, fracture as Mode-I separation) hold once a self-referential system takes up and redistributes load. Prevailing statistical evaluation practice treats such sequences as collections of independent features; by construction it cannot resolve the coupled material response (strain read along several axes of one $\sigma = E\varepsilon$ law) that SRDS measures.

A reversed burden of proof follows. We do not defend the mechanics case by case; it is the frame

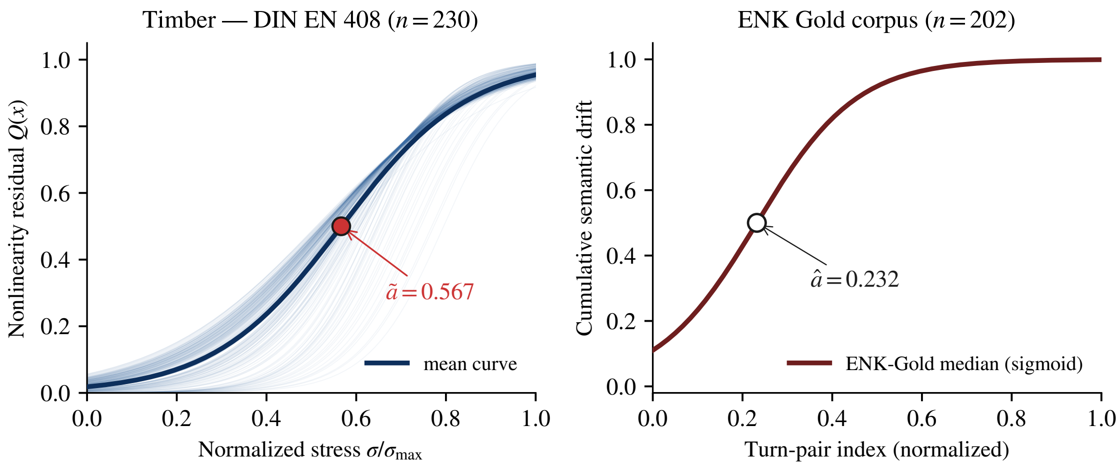


Figure 2: Nonlinearity residual $Q(x)$ of timber (spruce, $n = 230$, $\tilde{a} = 0.567$) and cumulative semantic drift of ENK-Gold ($n = 202$, $\hat{a} = 0.232$).

within which we measure. A falsification would have to exhibit a finding that *violates* the mechanics: a yield-point location that says nothing about strength; a truss that bends under axial load; a load accumulation that spontaneously reverses. These falsification tests are stated explicitly in §8.

The frame also delimits *what holds when*. Four physical axes of demarcation: (i) **forced vs. free sequence**: under machine-imposed ordering (tensile test) a path surrogate is physically meaningless, the existence test being model comparison; under free ordering (conversation, market, earthquake) the surrogate is mandatory; (ii) **uniaxial vs. multiaxial**: inflection and amplitude are two descriptors of *one* stress–strain response, coupled through $\sigma = E\varepsilon$, hence not separable quantities but the same material response read along two axes; (iii) **support topology**: it decides, before any measurement, which axis bears load and which falls into the null member (§5); (iv) **no universal constant** (no-go theorem, §2.5): what is shared is the *functional form*, not the numeric value. The axes are not auxiliary assumptions but consequences of the applied mechanics; they fix the scope of each test before the measurement.

1.3 The General Form

The question is not "How does one detect manipulation in a chat?", it is more general: what happens structurally when a self-referential system stands under load? Table 1 lists ten substrates across eight research fields (Figure 3; timber and ENK conversation compared directly in Figure 2): materials science (timber, metals, battery), geophysics (earthquakes), biochemistry (lignin), finance (SPY), democracy (V-Dem), oncology (colorectal carcinomas), language (LLM conversations), music (MetaMIDI). The cumulative response follows a sigmoid saturation function with well-defined inflection in each; the substrate-specific form within this class family (Logistic, Weibull, Gompertz, Hill) is resolved in §2.4. The carriers differ: the class family of the functional form is shared, the specific form within it is substrate-specific.

What must redistribute, deforms; what can no longer redistribute, breaks. This path-dependence of load-bearing capacity corresponds to the institutional path-dependence reading in historical institutional economics (North, 1990; Pierson, 2000; Mahoney, 2000). The structural-mathematical statement that recurs across ten substrates is not confined to conversation analysis; it was merely measured there first. Across twelve conversation corpora ($\approx$ 35,300 dialogues from 14 sources) the same mechanism replicates; ProsocialDialog serves as the pre-registered null control; Sotopia and Google COVID Mobility, pre-registered with directional positive hypotheses, were falsified and thereby map the scope conditions.

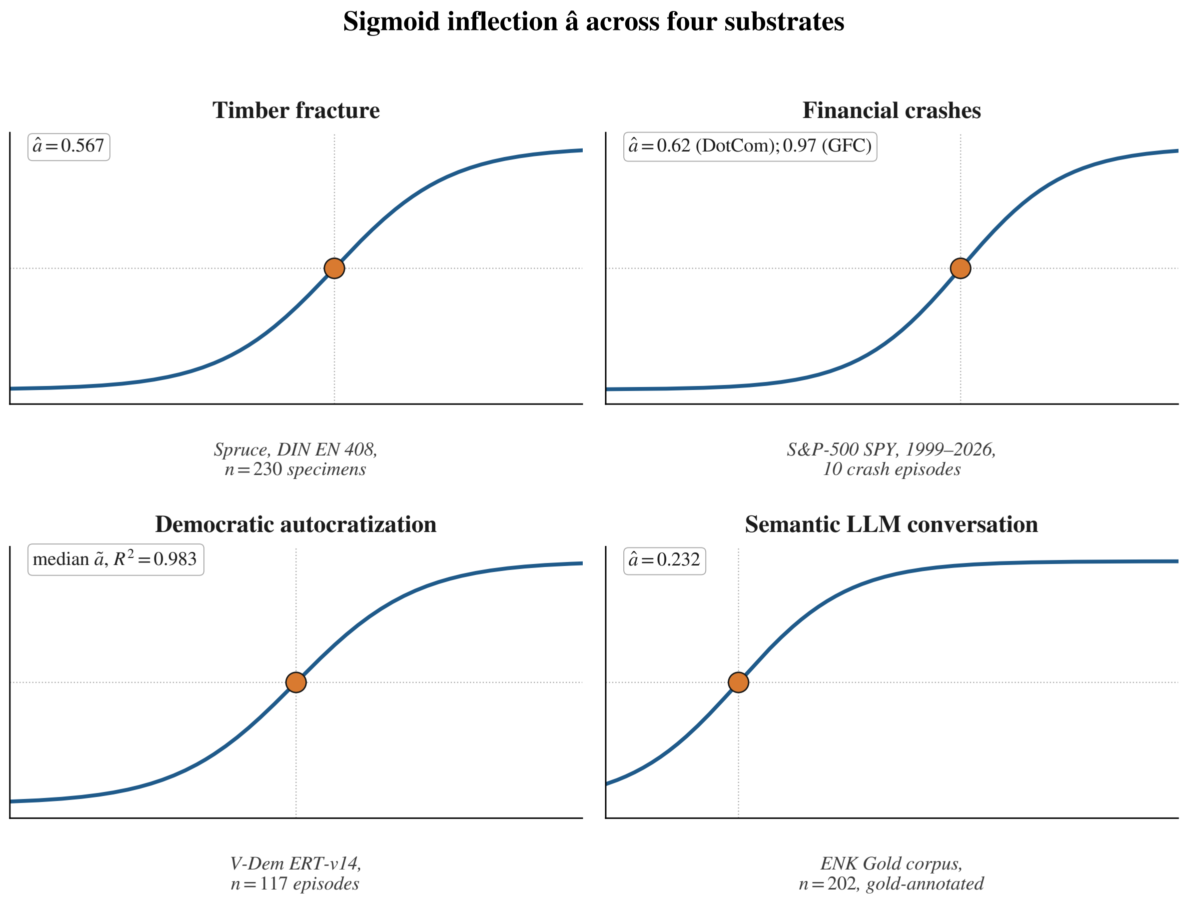


Figure 3: Sigmoid fit of cumulative response across four substrates: timber ($\tilde{a} = 0.567$), financial crashes ($\hat{a} = 0.62$ DotCom / 0.97 GFC), V-Dem autocratization (median $R^2 = 0.983$), LLM conversations ($\hat{a} = 0.232$). Full list in Table 1.

Table 1: Ten substrates, common sigmoid class family with substrate-specific normalization Π and form

Substrate	Π	n	Headline
Timber (*Picea abies*, DIN-EN-408)	$Q(x)$	230	$r(\hat{a}, \sigma_{\text{break}}) = -0.83$, $\tilde{a} = 0.567$
Multi-Material Multiaxial Fatigue (136 materials)	$\sigma(\varepsilon)$	914	$\hat{a}$ reads non-proportionality; accumulation in the ENK band (44–62% Chelton)
Finance SPY (1999–2026)	cumsum	10+	crash $R^2 = 0.973$ vs. calm 0.483
Lignin Kraft pulping (SP-LCC)	$\sigma(\varepsilon)$	72 cond.	$\rho_S = -0.78$ (Weibull, mode ii)
V-Dem ERT-v14 autocratization	cumsum	117 (89 fitted)	median $R^2 = 0.983$
Earthquakes USGS, 6 regions	cumsum	284 seq.	$\rho(\hat{a}, \text{mainshock}) = +0.673$ Bonferroni
Battery CALB (arXiv:2502.18807)	cumsum	26 cells	KS $D = 0.733$ ($p < 10^{-13}$); knee localization + G7 hysteresis
Colorectal TCGA	cumsum	579	MSI–MSS $\Delta\hat{a} = -0.153$, $p < 10^{-4}$
LLM conversations (ENK gold)	cumsum	202	$\rho = -0.359$ cluster-robust, ridge $\rho = 0.700$
Music (MetaMIDI + music21 + Tagtraum)	cumsum	2,840	6-genre $\tau = +1.000$, $p = 0.003$; median $R^2 = 0.975$

1.4 What This Paper Delivers

This paper delivers three contributions. (i) SRDS (Self-Referential Dissipative Systems) as a formal framework with five axioms A0–A4, three core theorems T1–T3, four aggregation modes (parallel-instantaneous, serial-cascaded, multiaxial, continuously-accumulating; substrates as mode mixtures, §2.4), and a no-go theorem (§2.5) against universal constants outside mirror symmetry, connected to formal mathematics (de Bruijn, 1968; Harrison et al., 2014; Mathlib Community, 2020). Full development and proofs are in the formal companion framework (Schessl, 2026b), the Lean-4 formalization in the companion `srds-enk-companion/lean4/PositivityProofs/`. (ii) ENK (Emergent Narrative Control, here operationalized as an embedding-geometry apparatus) as a special case for the conversation substrate: gold-annotated corpus $n = 202$, twelve external replication corpora, five measurement axes, cluster-robust inference at turn-pair level (cumulative family $\bar{\rho}_1 = 0.928$; Schessl, 2026a). (iii) Falsifiability via synthetic null specimens (1,000 per substrate, KS-test $p < 10^{-3}$ on all five substrates tested against AR(1)/RW/WN), a model comparison against the saturation class family (Gompertz/Weibull/Hill), three axiom-violation controls with *predicted* null signals, and Bonferroni-conservative reporting.

Under the conservative safety factor γ_M Bonferroni ($\gamma_M = \alpha/9 \approx 0.0056$; nine per-substrate tests, with multiaxial fatigue as the tenth substrate carrying separately on the accumulation axis; family and per-substrate test §8), six of the ten substrates are conformant (Timber, ENK Conversation, Earthquakes, Battery under caveat, Music with $\tau = +1.000$, V-Dem with sigmoid load-bearing capacity median $R^2 = 0.983$). Multi-Material Multiaxial Fatigue carries on the accumulation axis (autocorrelation structure in the ENK band, the Chelton inflation band of the ENK conversation substrate at 44–62%; $\hat{a}$ reads the load-path non-proportionality) with pooled-damped cross-material outcome coupling; three carry nominally with named caveats (Lignin, Colorectal, Finance); in addition, there are a pre-registered null control (ProsocialDialog) and two falsified positive predictions that map the scope conditions (Sotopia, Google COVID Mobility). The pattern is the finding: the signal appears *where the mechanics predicts it* and is absent *where the mechanics predicts its absence*. A predicted null is thus not a gap but a passed falsification test (§8).

Substrate-specific deviations of the inflection location are reported not as confirmation of a universal value but as the measured asymmetry of each carrier. Axiomatics, Lean-4 proofs, gold corpus, metric code, and cross-substrate raw data lie open (companion directory `srds-enk-companion/`); the operational calibration thresholds of the state machine remain withheld. For a deployed stability detector, publishing the exact trigger values would enable targeted evasion (responsible disclosure). The structural architecture, axioms, theorems, and Lean-4 proofs are fully published; no reported finding depends on the thresholds, since the structural load-bearing capacity is threshold-independent. Simon et al. (2026) has concurrently proposed a programmatically related discipline, *learning mechanics* (arXiv:2604.21691); it is the convergence anchor to which this work independently converges.

The following sections develop the formal framework (§2), the measurement apparatus (§3), the cross-domain findings (§4), the structural scale (§5), the cross-corpus methodology (§6), the external convergence (§7), and the falsification anchors (§8).

2 SRDS Framework and Mathematical Load-Bearing Capacity

2.1 Two Core Equations

The structural grammar of SRDS consists of two coupled recursions. The first describes the evolution of state under external excitation:

$$z_{t+1} = F_t(z_t, u_t).$$

The second describes the evolution of the operator that propagates the state:

$$F_{t+1} = \Phi(F_t, z_t, u_t).$$

The first equation is classical feedback structure (Wiener, 1948; Ashby, 1956). It alone does not suffice: a specimen remembers its loading history (Bauschinger effect), a conversation carries path-dependence. The second equation makes this memory explicit: the operator F_t is itself the state of a superordinate system that changes its own response function over the course of loading. A structural fixed point in the sense of autopoiesis (Maturana and Varela, 1980) lies where $\Phi(F^*, z^*, u^*) = F^*$.

2.2 Five Axioms A0–A4 + Four Regularity Conditions L1–L4

Table 2: Axioms of the SRDS class

#	Axiom	Form	Substrate analogy
A0	Self-reference	$F_{t+1} = \Phi(F_t, z_t, u_t)$	Material memory (Bauschinger)
A1	Dissipation	Lyapunov functional V: $V(F_{t+1}) \leq V(F_t)$ in expectation (Prigogine, 1980)	Heat export, local order through exported entropy
A2	Accumulation	Damage Σ monotone: $\Sigma(F_{t+1}) \geq \Sigma(F_t)$ (Kachanov, 1958; Lemaitre, 1985)	Damage parameter, crack-area fraction, budget accumulator
A3	Transition	$\sigma(\Sigma)$ possesses a unique inflection point $\hat{a}$ (sigmoid form)	Yield point, tipping point (Scheffer et al., 2009)
A4	Irreversibility	Trajectory with $\Sigma > \hat{a}$ returns with probability < 1 (Seifert, 2012)	Hysteresis, plastic deformation

Explicit reduction. A2, A3, A4 are, under regularity conditions L1–L4, theorems derived from A0+A1: L1 (Lyapunov semi-order), L2 (saturation bound), L3 (asymmetry measure in the sense of stochastic thermodynamics), L4 (return protocol + elasticity below yield). Concretely: A2 follows from A0+A1+L1+L2; A3 from A2+L2; A4 from A0+A1+L3 (unconditional) or +L4 (conditional) (reduction theorem, proved in full in the formal companion framework (Schessl, 2026b)). The reduction is made explicit so that the load-bearing capacity of each axiom can be tested per substrate, rather than assumed uniformly across all substrates. For the predicted null (ProsocialDialog) it is precisely this explicit reduction that explains the null finding (L1 violation, $\beta \to 0$, recovery balances damage); for Sotopia (A0, symmetry instead of asymmetric self-reference) and Google COVID Mobility (L3 / $\sigma \perp \varepsilon$, both time-driven) it explains *post-hoc* why pre-registered positive predictions were falsified and maps the scope conditions (§8).

2.3 Two-Scale Mapping + Renormalization Operator

SRDS separates two scales: material level (single specimen, load-bearing capacity on the sample) and structural level (system into which the specimen is built). The renormalization operator M links them:

$$M_{\ell \to \ell'} = N_{\ell'} \circ R_{\ell \to \ell'} \circ C_{\ell \to \ell'}.$$

C **coarse-graining:** micro-trajectories to blocks (windows, clusters, representative volumes). R rescale: effective parameters on the coarser scale follow parameter flow $\theta_\ell \mapsto \theta_{\ell'}$. N normalization: extracts dimensionless invariants so that the renormalized description sits on the same structural tier as the source description.

Five operative tiers carry the empirical work: token (ℓ_0, word embeddings) $\rightarrow$ turn-pair (ℓ_1, semantic velocity, frame drift) $\rightarrow$ chat (ℓ_2, $\hat{a}$, budget accumulator) $\rightarrow$ corpus (ℓ_3, $\hat{a}$ distribution) $\rightarrow$ society (ℓ_4, democracy index trajectories). Under the five axioms the dynamics on each scale admit a common normal form $\dot{z} = F(z, \Sigma) - \nabla_z \Pi(z, \Sigma) + \Theta \cdot \nabla_z I(z)$, form-invariant under M (Kadanoff-Wilson signature). Micro-details drop out under coarse-graining; the structural grammar remains.

2.4 Aggregation Modes and Mode Mixture

Under load the sigmoid form of a substrate follows one of four aggregation modes of the micro-imperfection cascade. Loading meets micro-imperfections, which activate in one of four ways: in parallel (CLT approximation), serially in a cascade (weakest-link statistics), multiaxially (hybrid), or continuously (diffusive averaging without a discrete failure event). The classical materials-science ductile/brittle typology (Considère, 1885; Tresca, 1864; Mohr, 1900; Bridgman, 1944) is a binary projection of this finer partition; SRDS distinguishes the modes through the sigmoid class family on $\Pi_S(X, Y)$, the image of the raw trajectory (X, Y) under the substrate-specific normalization Π from Table 1 into $[0, 1]^2$ (X: load or ordering variable, Y: response variable): Logistic, Weibull, Gompertz, and Hill are data-borne form classes on a single substrate, not the result of separate tests.

Table 3: Aggregation modes and form class

Mode	Activation	Form class	Typical $\tilde{a}$	Empirical anchor
(i) parallel-instantaneous	all micro-elements simultaneously under load, abrupt failure event	**Logistic** (CLT)	≈ 0.55	Logistic dominant in 6/8 material substrates of the model comparison (NR60 100%, FKM 80%, battery 100%; median $R^2 \geq 0.94$); ENK 100% across 12 conversation corpora
(ii) serial-cascaded	micro-failures cascade hierarchically	**Weibull** (weakest-link via order statistics)	≈ 0.30	CCP60 (cold-castable polyether, soft PE; public Dryad dataset): Weibull AIC-win (0% Logistic, median $R^2 = 0.96$, $n = 9$), $\rho(\hat{a}, \text{outcome}) = +0.867$: weakest-link failure of the polymer chain network
(iii) multiaxial	tensile and compressive components simultaneously	hybrid (Hill/Gompertz)	—	Multi-Material Multiaxial Fatigue ($n = 914$, 136 materials, Materials Cloud public): non-proportional multiaxial load; form partition as refinement
(iv) continuously-accumulating	gradual micro-defect accumulation without discrete failure event	**Gaussian/Logistic** (diffusive averaging)	≈ 0.55	Battery single-cell Logistic (CALB, arXiv:2502.18807, $n = 26$)

The binary creep/brittle-fracture reading remains useful as operating-range heuristic: creep $\approx$ modes (i) and (iv) with late inflection and slowly growing accumulation; brittle fracture $\approx$ mode (ii) or abruptly-activating mode-(i) substrates with late asymmetry, where the accumulation unloads abruptly instead of growing. The concrete substrate evidence distributes across both modes (financial crashes §4, conversational corpora §6). The sign-inversion between covert and overt substrates is itself a measurement value, not an artifact. Hysteresis appears in substrates with a long time axis: V-Dem ($n = 117$) shows 67% recovery within a decade but never a full return, and the fatigue shift $\rho \approx -0.30$ ($p = 0.029$) documents that $\hat{a}$ migrates path-dependently.

Mode mixture as substrate property. A substrate need not occupy a single mode but can be a linear combination of several; the substrate-specific characteristic is then the mixture vector $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3, \pi_4)$ on the mode simplex Δ_4. A focused load-case study measures this mixture for the conversation substrate under pushback/sycophancy loading (Schessl, 2026c). Mode identities (i)–(iv) remain universal: basis families in the Π-sigmoid class vector space; the renormalization operator R_η and the no-go theorem (§2.5) are not affected by mode mixture, the substrate-specific asymmetry now resides in the mixture vector as well as in the inflection location.

Where a substrate exhibits none of the four modes, the question is which axiom is violated, not whether the framework holds universally. Null findings (§8) follow from the same axioms as positive findings: both are direct consequences of the same reduction.

2.5 Mathematical Load-Bearing Capacity: Birkhoff-Hopf, No-Go, T1–T3

The mathematical load-bearing capacity of the framework rests on three building blocks. The first is the Birkhoff-Hopf contraction (projective contraction theory on monotone kernels; Birkhoff, 1957; Hopf, 1963; Seneta, 2006): For a Gaussian smoothing operator T_ε with kernel $\exp(-(x-y)^2/\varepsilon)$ on monotone profiles over $[0, 1]$ (Hilbert diameter $\Delta_\varepsilon = 2/\varepsilon$), in the limit $\varepsilon \to 0$:

$$1 - \kappa(\varepsilon) \sim 2\, e^{-1/\varepsilon}, \quad \varepsilon_n \asymp \frac{1}{\log n}.$$

The global contraction budget closes exponentially in $1/\varepsilon$, the local diffusive gap only linearly: local smoothing scales favorably, while global order grows disproportionately with span. This exponential degeneration form is the classical Hilbert-metric contraction bound of projective geometry; recent work on Sinkhorn iteration in continuous settings confirms this exponential form and sharpens the associated rates (Chizat et al., 2024; Eckstein, 2023, the latter for kernels of bounded growth not bounded away from zero). The $\varepsilon \to 0$ degeneration is thus a singular-limit phenomenon of the worst-case bound, not the operative convergence rate. Analytic proof (fixed-point existence, sharp decay, no-go) in the formal companion framework (Schessl, 2026b); formalized in Lean 4 in `srds-enk-companion/lean4/PositivityProofs/`.

No-Go Theorem. Let R_η be the damped renormalization operator on anchored monotone profiles, with TP_2-positive integral kernel and damping η toward a reference. If the entire damped operator (including the reference) commutes with the reflection $SQ(x) = 1 - Q(1 - x)$, the fixed point Q^* is unique, and its density has a unique non-degenerate maximum, then the inflection location is $\Sigma_c = 1/2$; mirror symmetry is thereby sufficient, not proved necessary. Conversely, in a reparametrization-closed class without such reparametrization-breaking structure no universal inflection constant exists; in one certified asymmetric direction the variable inflection location $\tau(\alpha)$ shifts continuously off the symmetric point (Σ_c denotes only the symmetry value $1/2$). The certified spectral bound $\rho(DR_\eta[Q^*]) < 0.45$ at the certified parameter point secures local geometric convergence; global geometric convergence follows from the separate sup-norm Banach bound. **Computer-assisted proof:** $\dot{Q}(1/2) \in [1.8412, 1.8963]$ via Arb certificate in interval arithmetic (256-bit, Krawczyk + Anselone + box enclosure, reproducible via `make verify`); Lean-4 code publicly accessible, certificate load-bearing per `#print axioms` with atomic axiom audit. The general statement for every admissible asymmetry and all (ε, η) remains open; the no-go theorem as a non-existence statement does not require it: a single certified asymmetric direction where the location shifts suffices. Substrate-specific asymmetry is thereby a measurement task, not a defect of the framework.

Substrate-specific scatter of the inflection. Substrate-specific asymmetry shifts $\hat{a}$ in structured fashion away from the symmetry value $1/2$, depending on how early the substrate yields under load; the deviations are not artifacts but the measurement value of the actual asymmetry of the respective substrate. Empirically: timber $\tilde{a} = 0.567$ (fiber asymmetry), battery $\tilde{a} = 0.646$ (CALB dataset), DotCom $\tilde{a} = 0.62$, GFC $\tilde{a} = 0.97$ (brittle-fracture asymmetry of a liquidity shock), earthquakes $\tilde{a} = 0.452$, lignin (boundary case: inflection at the origin, outside the interior $(0, 1)$, since the broad β-O-4 strength distribution has no endurance limit). Even sunflower phyllotaxis, the Fibonacci pattern par excellence, yields, across eight heads (floret-anthesis accumulation; Dryad 10.25338/B8963G), a mean inflection $\tilde{a} \approx 0.36$, not $1/\varphi$: $\hat{a}$ measures a yield transition, not a phyllotaxis angle. That several substrates fall near $1/\varphi \approx 0.618$ is a measurement reading, not a law; the structural reading of this proximity follows below.

Structure determines the value, family-internally and algebraically. The non-universality of the no-go has a constructive obverse: *within* a structured family the characteristic scale value is not fitted but algebraically determined, namely the Perron-Frobenius eigenvalue of the symmetry-specific linear substitution matrix (length factor λ; the prototile-count matrix carries λ^2). In aperiodic order (quasicrystals) this inflation eigenvalue is, in the standard irreducible cyclotomic cases, a Pisot number of degree $\phi_{\mathrm{Eul}}(n)/2$ (the symmetry fixes the degree, the substitution rule the value): five-fold the golden ratio φ (minimal polynomial $x^2 - x - 1$), eight-fold the silver ratio $1 + \sqrt{2}$ ($x^2 - 2x - 1$), twelve-fold $1 + \sqrt{3}$ ($x^2 - 2x - 2$), seven-fold the irreducibly cubic Perron-Frobenius root $\lambda_7 \approx 2.247$ of $x^3 - 2x^2 - x + 1$ that breaks the quadratic case (Pautze, 2017; Nakakura et al., 2019; Hare et al., 2018). The value $1/\varphi \approx 0.618$ is the inverse scale of the five-fold golden eigenvalue φ (whose algebraic conjugate is $-1/\varphi$), *one* of this spectrum, not *the* universal constant. Across *genuinely disjoint* substrates that share no common structural parameter, no common structure-to-value law exists; only the form is universal. Aperiodic order is the test bench here, not a substrate of this study: the one field where "structure determines the value" can be checked against an exactly known algebraic ground truth, preregistered and tested (Figure 4). This sharpens the no-go twofold: no universal *value* and no universal *law* across disjoint substrates, alongside hard family-internal determination.

Three core theorems T1–T3.

#	Theorem	Statement
T1	Information-stability correspondence	$\mathrm{sign}(\mathrm{slope}_W I) = -\mathrm{sign}(\mathrm{Tr}\, A)$: over a finite window the mutual-information trend signs the conditional phase-space volume rate $\mathrm{Tr}\, A = \sum_i \lambda_i$ (Lyapunov-exponent sum); the $\lambda_{\max}$ reading is the dominant-mode case
T2	Autopoietic stationarity	$\mathcal{F}(\mathcal{F}) = \mathcal{F}$ yields non-equilibrium steady state with $\dot{D} = 0$ (entropy production = dissipative export)
T3	Goldilocks principle	Optimal function in a neighborhood of the edges of criticality, not at the measure-zero point of exact criticality

T1 connects to the stochastic thermodynamics of differentiable systems. There, in these differentiable settings, dissipated entropy is coupled to phase-space contraction (the sum of Lyapunov exponents), and the entropy-flow rate is spectrally bounded by the extremal eigenvalues of the symmetric Jacobian part (Das and Green, 2025): the same sign-coupling between the phase-space volume rate (the Lyapunov-exponent sum) and the information/entropy flow that T1 posits (the $\lambda_{\max}$ reading being the dominant-mode case). An exact identity linking entropy production and mutual information for overdamped Langevin dynamics (Cho and Jeong, 2025, preprint) further supports the direction. These external results are convergence anchors, not load-bearing premises. The full proof of T1 stands on its

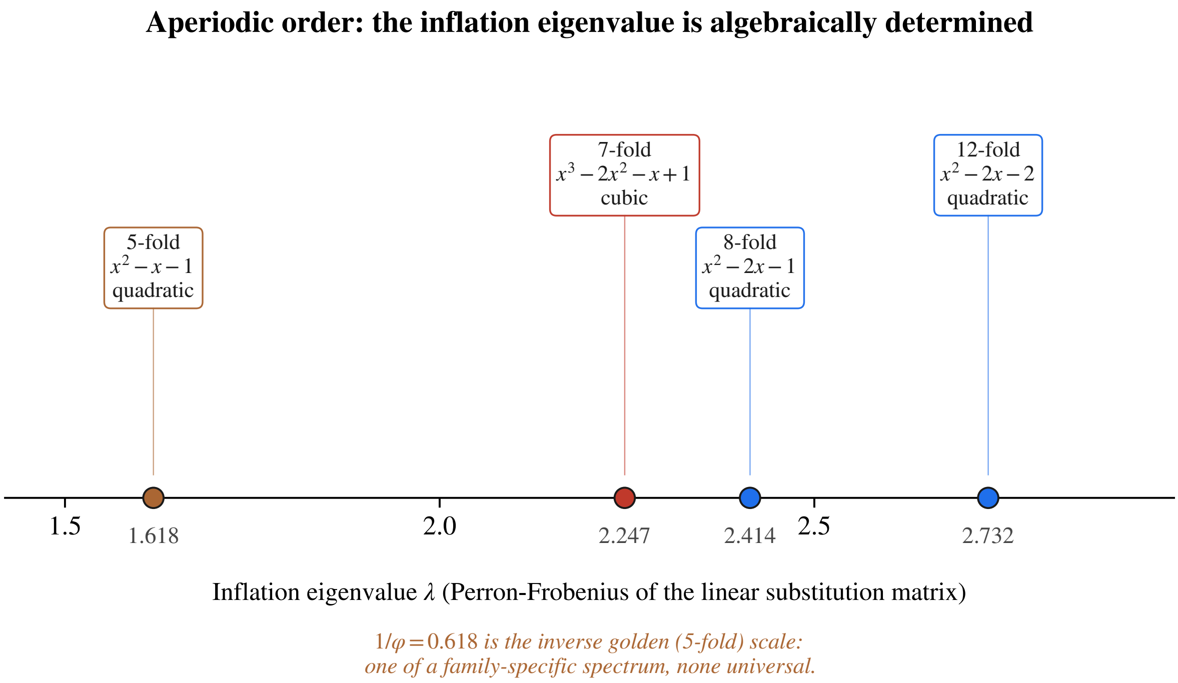


Figure 4: Aperiodic order as the constructive counterpart to the no-go: the inflation eigenvalue (Perron-Frobenius of the symmetry-specific linear substitution matrix) is an algebraically determined Pisot number of degree $\phi_{\mathrm{Eul}}(n)/2$. Five-fold golden φ, eight-fold silver $1+\sqrt{2}$, twelve-fold $1+\sqrt{3}$ (quadratic); seven-fold irreducibly cubic (x^3-2x^2-x+1, breaks the quadratic case). The value $1/\varphi \approx 0.618$ is the inverse golden scale (algebraic conjugate of φ: $-1/\varphi$), one of a family-specific spectrum. Pure aperiodic order, no SRDS substrates.

own in the formal companion framework (Schessl, 2026b).

Three further works realize the same structure in mathematically disjoint formal languages: AdS-GNN (Zhdanov et al., 2025, conformal-equivariant operator geometry), Topological Deep Learning (Schmidt et al., 2026, topological vs. combinatorial generalization), mesh-agnostic operator composition (Yang et al., 2026). The class is consistently realized across three mathematical languages: functor-invariant, not number-faithful. Functor invariance here is a functorial realization in the category-theoretic framing (Mac Lane, 1971; Spivak, 2014).

A functor $\Psi : \mathrm{SRDS} \to \text{Aggregate-Class}$ realizes the sigmoid form with substrate-specific inflection (functorial realization; full development and proofs in the formal companion framework (Schessl, 2026b)).

3 Measurement Apparatus ENK (Material Level: Conversation as Specimen)

3.1 Embedding Geometry as Measurement System

Conversations are measured with transformer embeddings: high-dimensional vectors in which related utterances lie close to each other. Sentence-transformers (Reimers and Gurevych, 2019; Mikolov et al., 2013 as token precursor) measure semantic similarity at sentence level. The primary encoder is all-MiniLM-L6-v2 (384 dimensions); all-mpnet-base-v2 (768 dimensions) provides the triangulation. Each turn-pair (u_t, a_t) is a measurement point; the ordered sequence along a conversation is the deformation curve of this specimen. Decreasing embedding variance *is* stiffening of the semantic space (a reduced degree of freedom); irreversibly growing cumulative displacement *is* plastic deformation of the specimen. The apparatus measures not a speaker's intent but the specimen's trajectory under load.

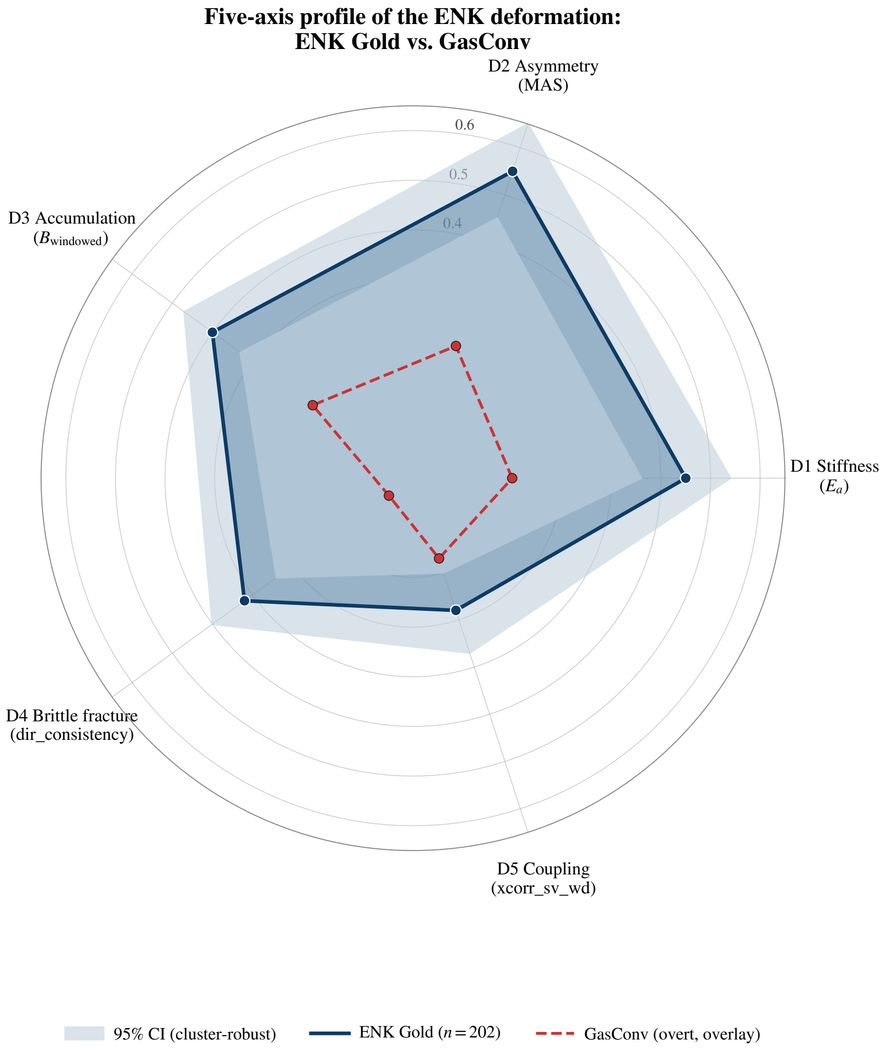


Figure 5: Five-axis radar of the ENK deformation profile (D1 stiffness / D2 asymmetry / D3 accumulation / D4 brittle fracture / D5 coupling) with cluster-robust confidence intervals, $n = 202$.

Two robustness anchors. First: format ablation (Markdown / code blocks / lists removed before recomputation of E_a) shows that 95% of the signal stems from semantic content, 5% from formatting ($n = 202$). Second: partial Spearman correlation against conversation length holds the signal strength stable ($\rho_{\text{partial}} = 0.485$ after controlling for $\log(\text{turn count})$). Length is, in the SRDS sense, loading duration, not a nuisance term: longer specimens accumulate more damage, as expected in any fatigue process.

3.2 Five Effective Dimensions: the Anti-Sparse Rule

The five measurement axes each capture a distinct component of the deformation response.

Table 4: Five effective dimensions of conversational deformation

Axis	Interpretation	Material analogue	Strongest metric	ρ (vs manipulation)
D1 Stiffness	Variance decrease as reduced degree of freedom	Strain hardening	$E_a = 1/\mathrm{Var}(\text{sem velocity})$	**+0.543***** (space-robust: MiniLM +0.543, MPNet +0.507)
D2 Asymmetry	User-AI imbalance in embedding space	One-sided load introduction (cantilever)	MAS (Meta-Asymmetry)	**+0.443*****, ICC=0.954
D3 Accumulation	Irreversible cumulative deformation	Creep / fatigue	B_w (windowed budget)	**+0.249**** cluster-robust
D4 Brittle fracture	Abrupt break-out from directional flow	Brittle fracture	dir_consistency	**-0.377***** cluster-robust
D5 Coupling	Dynamics-structure covariation	Cross-field coupling	xcorr(sv, wd)	**+0.210****

Plus core construct $\hat{a}$ (sigmoid inflection on cumulative damage over cumulative strain): $\rho = -0.359^{***}$ strain axis, Cross-Corpus 4/5 (Awry marginal). (Reported here is the correlation $\rho(\hat{a}, \text{mal-frac})$; the inflection location $\hat{a}$ itself is constructed substrate-specifically, §2.5.)

Anti-sparse rule. The five axes are coupled, not redundant. Hooke's law $\sigma = E\varepsilon$ couples stress, stiffness, and strain analytically: the same coupling reappears in the ENK axes. D1 (stiffness) and D3 (accumulation) are two projections of the same response. Lasso or other L1 procedures would discard three of the five axes under multicollinearity penalties. But the axes carry the same material response from orthogonal projections, so discarding them loses information, not noise. This coupling is an analytic constraint of the linear-response (Hooke) regime of the dissipative dynamics (A0+A1), not statistical multicollinearity one could regularize away. That is the wrong paradigm: materials testing, not regression statistics over independent features. Autocorrelation is signal, not a nuisance term: measurement points on a stress-strain curve *are* correlated (lag-1 $\bar{\rho}_1 = 0.928$ cumulative family, see §3.3).

3.3 Gold Reference Corpus + Cluster-Robust Inference

$n = 202$ conversations were annotated, 11,639 turn-pairs, across four LLM families (ChatGPT / Claude / DeepSeek / Gemini) and five contributors. A structured reference-measurement protocol (Claude Opus 4.6) assigns 17 annotation fields to each turn-pair (chunk-overlap consistency $\alpha = 0.87$; parallel run with independent LLM: 93.2% binary label agreement). Manipulation incidence: 42% manipulative turn-pairs. Four-phase classification P1 baseline $\rightarrow$ P2 buildup $\rightarrow$ P3 TypX (manipulation inflection) $\rightarrow$ P4 consolidation. Construct reliability rests not on a single annotator cohort but on cross-corpus replication (§6, §8): the same signal across twelve independently collected corpora demonstrates reliability more dependably than a single inter-rater study.

Cluster-robust inference. Lag-1 autocorrelation of the cumulative accumulator family: mean $\bar{\rho}_1 = 0.928$, individual metrics up to $\rho = 0.944$. Block bootstrap (2,000 iterations, full conversations as blocks) restores the specimen as analysis unit. Cluster-robust block bootstrap corresponds to the cluster-correction method in econometrics (Cameron and Miller, 2015; Wooldridge, 2003): sequential measurements along a deformation curve are not independent, and naive pooled inference overstates the confidence. Under cluster-robust inference, 47 of 113 pooled-significant metric-target associations remain conformant at the safety factor γ_M; the remaining 66 are not identifiable under the correct inference architecture. The pooled p-values overstate confidence by 42% corpus-wide, and by about 55% for the cumulative family (Schessl, 2026a, Eq. 10). The safety factor itself is defined over the nine per-substrate tests ($\gamma_M = \alpha/9$, §1); the association count here uses γ_M as a conservative per-association threshold and does not form a separate Bonferroni family over the 113 associations.

Multivariate chat-level ridge (cluster-robust block bootstrap). Ridge regression on 15 chat-wide summary quantities reaches $\rho = 0.700 \pm 0.076$ against the continuous manipulation degree. Lasso reaches $\rho = 0.686 \pm 0.073$, comparable, consistent with the anti-sparse rule.

LOPO batch diagnostics. Leave-one-person-out: macro-$\rho = 0.339$ vs length reference 0.174 ($\Delta = +0.165$); pooled $\rho = 0.552$ vs 0.410 ($\Delta = +0.142$, with autocorrelation contribution). LOPO is not a model-generalization metric but a batch-invariance test: does the measurement system carry across the identity of contributors? $\Delta > 0$ indicates substrate-independent signal. An underpowered batch is therefore an expandable measurement (§8.5), not a falsification: the signal exists, its sharpness grows with batch diversity.

3.4 Crystallization Signature: Stiffening as Damage

Three-state classification via three material analogues: h_{bond} (semantic bond decay rate between turns, analogous to H-bond decay in polymer mechanics), hydrophobic packing density (frequency of dense topic clusters in the embedding space), and mean semantic velocity (drift rate of the embedding trajectory). Classification by median split: *crystalline* (h_{bond} and packing both above median), *glassy* (h_{bond} below median AND semantic velocity below 25th percentile), *amorphous* (remainder).

Table 5: Manipulation rate per state ($n = 196$)

State	n	Manipulation rate	Binomial p (against base rate 42%)
crystalline	52	**71.2%**	0.0020
glassy	42	26.2%	0.0012
amorphous	102	44.1%	n.s.

$\chi^2 = 18.813$, $p = 0.0001$, Cramér's $V = 0.310$, cluster-robust 95% CI [0.198, 0.377]. Pairwise Mann-Whitney crystalline vs glassy: $U = 1650$, $p < 0.0001$, $r_{rb} = -0.511$ (large effect).

Cross-corpus confirmation. h_{bond} decay rate inverts sign between covert and overt manipulation: ENK-Gold $\rho = +0.398^{***}$ (crystalline stiffening), GasConv $\rho = -0.250$ (glassy disruption). Fisher meta over eight external corpora: $\chi^2 = 68.96$, $p < 10^{-6}$. Manipulation stiffens the conversational space; it does not randomize it. This is compatible with neural collapse (Papyan et al., 2020), but at conversation level rather than representation level.

3.5 Structural Axes: What the Five Main Axes Bear

The five main axes are best-of-class of a broader bundle of validated measurement concepts. The full inventory carries 38 cluster-robust concepts out of 77 (49%; full inventory in companion work in preparation).

The selection of load-bearing sub-axes follows the structural-engineering measurement paradigms: load combinations, fracture mechanics, stability, multiaxial load response, bond hierarchy, and microstructure.

Load combinations. Eurocode load combinations (corresponds to the EN 1990 load combination convention $\eta = E_d/R_d$ in structural design; Eurocode 0). Permanent load $G_{\text{baseline\_sv}} = -0.478^{*}$ (favorable initial configuration carries through); variable load $Q_{\text{mean\_sv}} = +0.531^{*}$ (strongest Q metric); utilization ratio $\eta = E_d/R_d$ via $\sigma_{\text{user}}/(B+1)$: $\rho = -0.403^{***}$.

Fracture mechanics. stress_conc (K_I): $\rho = -0.312$ hold-out replicates; K_{IC} v2: $\rho = -0.247^{*}$; Griffith energy balance turn-level: discriminability 0.878** (critical load index 6.78, separation value from the cross-corpus spread, not an operative trigger).

Stability: creep, not buckling. The Euler slenderness ratio $\lambda = n_{\text{turns}} \cdot \beta/\text{WD}$ (Euler, 1744) is, on this substrate, almost entirely determined by sequence length ($\rho(\lambda, n_{\text{turns}}) = 0.94$–$0.99$) and sign-labile after length control; elastic buckling (stability loss, second-order theory) is not the governing failure criterion here; as a second-order limit state it is examined separately (companion). The governing mode is creep: uniform dynamics correlates robustly with load after length control ($\rho_{\text{partial}}(\mathrm{CV}_{sv}) = -0.36$, cluster-robust). Hall-Petch segment granularity (corresponds to the grain-size strength relation $\sigma_y = \sigma_0 + k/\sqrt{d}$; Hall, 1951; Petch, 1953): $\rho = +0.381^{*}$ inverted. Avrami kinetics of stiffening (corresponds to the phase-transformation kinetics in metallurgy; Avrami, 1939; Kolmogorov, 1937): double-logarithm plot shows two regimes with kink at mid sigmoid phase, replicated across seven corpora. A convergent result: Aguilera et al. (2025) report explosive phase transitions in neural networks on curved statistical manifolds with a hysteresis signature. This independently supports, from a statistical-physics substrate, the sigmoid-inflection form measured here.

Multiaxial load response. Mohr deviatoric $|\text{asymm}| = -0.246$, partial -0.259^{***} after B_w control (suppressor effect). Shear stress $G_{\text{shear}} = \text{DIV}/|\Delta\text{WD}|$: $\rho = +0.292$ cluster-robust. Second principal stress σ_2 via MAS, orthogonal to $|\text{asymm}|$ ($\rho = 0.042$).

Bond hierarchy (protein analogy). H-bond decay $\rho = +0.398^{***}$ (LOPO 6/6); hydrophobic packing $\rho = +0.230$ cluster-robust; disulfide (long-range references) $\rho = +0.215$ in 5/6 LOPO. Weak bonds react 1.4$\times$ more strongly than strong ones, the same weak-vs-strong-bond dichotomy as §5.4 lignin and §3.4 crystallization.

Microstructure. Grain-boundary NC2 (Newman conductance on 2-cluster partition, cf. Newman and Girvan, 2004; not to be confused with Neural Collapse Papyan et al., 2020): $r = 0.997$ across encoder spaces (space-robust). PCA isotropy: $\rho = -0.256^{*}$. Manipulation is more isotropic than nominal conversation: the semantic space loses its preferred directions.

Cross-scale bridge. $h_{\text{bond\_decay}}$ (token scale, S2) $\rightarrow$ stress_conc (chat scale, S4) partial $\rho = -0.573^{*}$ across 5/5 persons (controlled for mal_frac and length). Strongest cross-scale evidence of the framework: what the token scale registers as bond decay carries through, structurally consistent, to chat-scale stress intensity.

Minimum sampling. $n_{\text{turns}} \geq 15$ as minimum loading duration for stable E_a determination, analogous to the Nyquist minimum-sampling theorem (sample-size lower bound for reconstructable inflection, not a frequency argument).

3.6 Comparison with Message-Level Detectors

Existing AI-safety systems operate message-wise (LlamaGuard (Inan et al., 2023), Perspective API, Detoxify (Hanu and Unitary team, 2020)). Head-to-head subtle manipulations ($n = 8$): SRDS 5/8, LlamaGuard 2/8, one tie. On ProsocialDialog ($n = 5{,}000$ raw conversations, of which 308 with $n_{\text{turns}} \geq 15$ per §3.5 minimum sampling) SRDS monitoring delivers $\rho = -0.008$ (zero false alarms; budget NORMAL in 307/308 specimens); LlamaGuard shows on the same data $\rho = +0.327^{***}$: systematic false alarms at therapeutic intensity.

A fire alarm measures acute, localized danger; a structural monitor measures cumulative load-bearing capacity. Both are needed, neither replaces the other. The complementary profile (SRDS dominant on covert cumulative deformation, message-wise detectors dominant on acute single violations) is the operational signature the framework predicts. The operational calibration of the state machine is described in §5 as architecture.

4 Cross-Domain: Ten Substrates, One Functional Form

Timber is the methodological anchor, not a singular case. The transfer to nine further substrates tests whether the *method* carries over: whether it adapts to a new carrier, not whether a particular number recurs. What is universal is the functional form of inflection; what is substrate-specific is the asymmetry of the carrier on which the load is redistributed (no-go theorem, §2.5). The normalization Π differs per substrate (nonlinearity residual for timber, $\sigma(\varepsilon)$ for metals, cumsum for sequences; §2.4): what recurs is the form class, not the numeric value. Each sub-section states the finding in one sentence and, where the mechanics predicts it, the *counterpart* at which the signal must be absent. The substrates support one another via shared mechanical axes (§4.13). Raw data in the companion `srds-enk-companion/`.

4.1 Timber under Tensile Load: DIN EN 408 (Anchor)

Spruce specimens in bone-shaped geometry per DIN 52188:1979-05, fiber-parallel under tension per DIN EN 408:2012-10 (Schessl, 2025). Across $n = 230$ specimens (sorting classes S7+S10+S13, balanced from $n = 319$): $r(\hat{a}, \sigma_{\text{break}}) = -0.83$, $p < 10^{-30}$ (Spearman $\rho = -0.99$). The construction itself carries this coupling: the absolute stress at the inflection, $\sigma_{\text{onset}} = \hat{a}\,\sigma_{\max}$, follows the onset-strength line $\sigma_{\text{onset}} = 5.27 + 0.433\,\sigma_{\max}$ ($R^2 = 0.9925$), and since the inflection axis is normalized by $\sigma_{\max}$, $\hat{a} \approx 0.433 + 5.27/\sigma_{\max}$ is, up to residuals, a decreasing function of strength; the line alone forces Spearman -1.0, so the near-perfect rank order follows arithmetically from this relation (decomposition in `analysis/timber_normalization_control.py`, companion). A permutation control (onset permuted independently of strength: $r \approx -0.74$) answers only the narrower question of whether the onset is independent of strength; it is not: it spreads (CV $= 0.228$), scales with strength, and the coupling holds within each grading class ($r = -0.83$ S7, -0.86 S10, -0.95 S13). The anchor's role is therefore instrument validation against known sorting physics (§1.1: the relation is established in timber engineering); this substrate does not evidence a mechanism beyond that established relation. The class-specific inflection means decrease monotonically with strength: S7 $\tilde{a} = 0.574 \pm 0.057$ ($n = 144$), S10 0.557 ± 0.043 ($n = 71$), S13 0.539 ± 0.026 ($n = 15$): the best class (knot-poor) carries the lowest $\tilde{a}$, consistent with $r < 0$. $\tilde{a} = 0.567$ lies measurably away from the mirror-symmetry value $\Sigma_c = 0.5$ (no-go) and measures the fiber anisotropy.

The counterpart sits within the substrate itself (§8): the near-elastic cohort ($R^2 = 0.994$, no plastic regime) carries **no** yield signal: the plastic regime *gates* the signal.

4.2 Multi-Material Multiaxial Fatigue: $\hat{a}$ Reads the Non-Proportionality

$n = 914$ strain-controlled specimens from the Multiaxial Fatigue Deep Learning Dataset (Chen et al., 2024; 136 material entries across 40 base alloys; AZ31B-Mg, pureTi, 16MnR steel, 6061/7075-Al); 31 material entries with $n \geq 8$ for material-wise S-N sigmoid fits. The input per specimen is a closed cyclic load path, a **hysteresis loop** in stress space. $\hat{a}$ reads its **load-path non-proportionality**, the *established* damage driver in multiaxial fatigue (critical-plane / Fatemi-Socie): out-of-phase loading rotates the principal stress axes, activates more slip systems, raises the dissipated energy per cycle (Palmgren-Miner). The autocorrelation structure lies, at Chelton 49.7%, in the same band as the ENK conversation substrate (PASS); intra-material $\hat{a}$ carries up to $|r| = 0.82$ (PA38-T6 $+0.82$, CuZn37 -0.74, CP-Ti -0.69); the sigmoid form is robust (74% $R^2 \geq 0.5$). $\hat{a}$ and the amplitude (VM amplitude $\rho = -0.717$) are coupled descriptors of *one* response ($\sigma = E\varepsilon$, §1.1b). Aggregated cross-material, $\hat{a} \leftrightarrow N_f$ is directionally correct ($\rho = -0.058$: earlier inflection $\rightarrow$ longer life) but **pooled-damped** across 40 materials $\times$ 48 load paths, a property of heterogeneous pooling, not a failure.

4.3 Finance SPY: Crash S-Curve against Calm Random-Walk

Daily returns of the S&P-500 ETF SPY over 27 years (1999–2026). Per episode, a sigmoid fits the cumulative drawdown, read as a deformation curve. The load-bearing finding is the **counterpart**: mean $R^2 = 0.973$ during crash episodes versus 0.483 in length-matched calm phases (Mann-Whitney $p = 0.0001$): calm phases are random-walk, not S-curve. This is the core negative control against "everything saturates": the sigmoid form is episode-selective, not an artifact. The severe crashes distribute across both modes; the poles are marked by GFC 2008 ($\hat{a} = 0.97$, brittle fracture) and DotCom 2000 ($\hat{a} = 0.62$, creep). After DotCom a permanent redistribution remains in the daily-return autocorrelation (hysteresis).

4.4 Lignin: Material-Class$\rightarrow\hat{a}$ via the Bond Hierarchy

Hydrothermal treatment degrades lignin via hydrolysis. We plot β-O-4 concentration against the P-factor (time-temperature severity) over $n = 72$ process conditions (95 samples, SP-LCC dataset, Alopaeus et al., 2025). Weibull sigmoid (serial-cascaded, mode (ii), weakest-link failure): $\rho_S = -0.78$. The β-O-4 bonds form a *broad* strength distribution (phenolic linkages cleave well before the non-phenolic ones), so damage is concave from the origin ($m < 1$): **no endurance limit**, the weakest ether bonds yield at the slightest load (Wöhler fatigue side, Wöhler, 1870; Palmgren-Miner accumulation, §5). The inflection therefore lies below the tested severity range; an interior $\hat{a}$ is not defined on these data and is not reported as a number, independently confirmed via birch AqSO (Schlee et al., 2023: β-O-4 already strongly reduced at the mildest severity, $\approx 34/100$ Ar) and kraft softwood (Mattsson et al., 2017: reactive β-O-4 cleavage early in the cook). Weak bonds (β-O-4, 60 kcal/mol or 250 kJ/mol) yield before the strong ones (5-5 linkages): a weakest-link failure that repeats, on the chemical level, the bond hierarchy from §3.5 and the crystallization signature from §3.4. It is the same **material-class**$\rightarrow\hat{a}$ structure as timber defect class and music genre (§4.13); lignin marks the limiting case of this axis, where the broad defect distribution pushes the interior yield point to the origin.

The design is cross-sectional and the data provenance is open; the material-class$\rightarrow\hat{a}$ finding is nominal under γ_M to that extent.

4.5 V-Dem: Autocratization Trajectories (Two-Span Beam)

ERT-v14 (Episodes of Regime Transformation, Maerz et al., 2024; V-Dem-based, Coppedge et al., 2024): $n = 117$ autocratization episodes. The trajectories are sigmoidal, median $R^2 = 0.983$ across the 89 fitted episodes; in 60 the inflection $\hat{a}$ falls within the episode's observation window, in the remaining 29 the fit extrapolates it beyond the window (16 past the episode end, where the transition is not yet reached; 13 before the episode start). $\hat{a}$ partitions into creep erosion ($\hat{a} < 0.5$; Hungary, Austria 1933) and sudden regime switch ($\hat{a} > 0.8$; India 1975, Czechoslovakia 1938). On the institutional scale, hysteresis appears: about two-thirds of states return within a decade to the pre-episode corridor, never fully (Wilcoxon $p < 10^{-4}$), the same loading-unloading asymmetry as battery and multiaxial fatigue (G7, §4.13). For states with multiple episodes, $\hat{a}$ shifts to earlier values with each subsequent one ($\rho \approx -0.30$, $p = 0.029$): the institution carries prior damage as material memory. The structural mechanics of these trajectories (support topology, truss buffering) follows in §5.

4.6 Political Speeches: Support Topology, Not an Effect Claim

Political speeches test the support topology. Bundestag speeches $n = 6{,}874$, National Socialist (NS) speeches $n = 11$, ParlaMint pilot HU+PL+RS+SI. The Bundestag null ($\hat{a}$, E_a without significant correlation) is the **predicted confirmation** of support topology. The speech corpus over the order of business is a truss, and a truss bears no bending: the load is *passed on*, not locally deformed (§5.2). This makes no claim about the effect of speeches, only about the chosen measurement scale. That the discourse load *reaches* society is shown by the $\ell_3 \rightarrow \ell_4$ force transfer (ParlaMint), **directionally confirmed**: the $\hat{a}$-discourse axis is negative across all four clusters (HU -0.40, PL -0.52, RS -0.21, SI -0.40; BCa-CI$_{95}$ $[-0.34, -0.15]$ excludes zero), the NC2 geometry axis $\rho = -0.34$ (BCa-CI$_{95}$ $[-0.48, -0.09]$, $p = 0.006$). The permutation $p = 0.207$ is starved of tail resolution at $G = 4$ (only 4! country permutations) and not load-bearing; the dependable statistic is the BCa-CI plus sign convergence across four independent clusters. NS speeches ($n = 11$) are illustrative (direction, not magnitude).

4.7 Earthquakes USGS: $\hat{a}$ Localizes the Mainshock

USGS-FDSN across six regions, $M \geq 2.5$ (2016–2021), 11,505 events $\rightarrow n = 284$ sequences (spatio-temporal clustering). Per sequence, a sigmoid fits the normalized cumulative energy release: median $R^2 = 0.965$, $\tilde{a} = 0.452$ ($\sigma = 0.199$). Against a null model with identical event statistics, KS-$D = 0.178$, $p < 0.001$: real inflection structure, not a random curve. The central statement: $\hat{a}$ **localizes the seismic transition**, $\rho_S(\hat{a}, \text{mainshock position}) = +0.673$, $p < 0.0001$ (Bonferroni): early $\hat{a}$ mark foreshock-dominated, late $\hat{a}$ mainshock-dominated sequences. The transition *localization* is strong; the magnitude *prediction* is weaker ($\rho_S(\hat{a}, \text{max magnitude}) = -0.176$, $p = 0.006$).

4.8 Battery Degradation, CALB (real): Knee + Hysteresis Accumulation

$n = 26$ CALB cells from the BatteryLife corpus (Tan et al., 2025) (0/35/45 °C). $\hat{a}$ localizes the knee point: the empirical $\hat{a}$ distribution is strongly shifted against the synthetic null: KS $D = 0.733$, $p < 10^{-13}$ (mean 0.667 vs. null 0.541; $\tilde{a} = 0.646$). Two interlocking findings: (i) the voltage hysteresis $V_C - V_D = 0.389$ V **grows with ageing**, $\rho(V_C–V_D, \text{cycle}) = +0.886$ over 92% of cells: irreversible, accumulating polarization, the same G7/Bauschinger mechanics as multiaxial fatigue and V-Dem (§4.13); (ii) the outcome coupling is mode-dependent and open ($\rho_S(\hat{a}, \text{EOL}) = +0.18$, n.s.; right-censored, only 7/26 cells reached 80% EOL). As with earthquakes, $\hat{a}$ localizes the failure knee reliably but does not predict lifetime strength.

4.9 Colorectal Carcinomas: TCGA (Material-Class→$\hat{a}$)

TCGA Pan-Cancer Atlas *coadread_tcga_pan_can_atlas_2018* (Hoadley et al., 2018): $n = 579$, of which 76 with microsatellite instability (MSI; MANTIS > 0.4) and 503 stable (MSS). Read along the **load axis (stage)**, the more damage-prone class reaches its yield point earlier: stage-aggregated MSI $\hat{a} = 0.286$ vs. MSS 0.439 ($\Delta\hat{a} = -0.153$), corroborated by per-patient stage ordering ($\Delta\hat{a} = -0.10$), robust across two orderings and isomorphic to the timber defect-class hierarchy S7→S13; the pre-specified sign is negative. Sorting *within* a stage by damage itself (mutation count), by contrast, flips the sign ($\Delta\hat{a} = +0.090$), a count-sorting artifact, not a yield signal. Mean $R^2 \geq 0.96$; bootstrap MSI 0.567 ± 0.101 vs. MSS 0.534 ± 0.038, KS-$D = 0.350$, $p < 10^{-4}$; the KS test establishes the distributional *difference*. The per-patient bootstrap means lie at $\Delta = +0.033$, opposite to the stage-aggregated headline (-0.153): the sign direction is aggregation-dependent.

The coupling of $\hat{a}$ to survival remains open given the small MSI cohort; the material law along the load axis is robust.

4.10 LLM Conversations (ENK Gold)

$n = 202$ gold-annotated conversations, $11{,}639$ turn pairs, four LLM families, five contributors. Univariate $\rho(\hat{a}, \text{mal-frac}) = -0.359$ cluster-robust (strain axis); multivariate chat-level ridge $\rho = 0.700 \pm 0.076$ (§3.3). Cross-corpus: $\hat{a}$ carries across the manipulation-corpus specimen classes (GasConv, LegalCon, MentalManip, P4G; Awry marginal), strongest LegalCon ($\rho = -0.751$); the coupling strength depends on which strain axis is read (canonical cumulative strain §2.5 or the FID axis, named per corpus); 47/113 associations cluster-robust at γ_M. The counterpart is path destruction: destroying the turn ordering (surrogate) lets $\hat{a} \leftrightarrow$mal fall into the generic cloud. Empirically, the real path stands out with $z = -4.86$ (§8).

4.11 Music: MetaMIDI (ℓ_2), Pre-Registered + Cross-Method-Validated

Pre-registered on 16 May 2026 (hypotheses H1–H3, genre order per Di Marco fixed before analysis, with a built-in shuffle falsification; the pre-registration is public in the companion, `data/musik_companion/`), run 17 May 2026. $n = 2{,}840$ MIDI pieces: 1,974 from music21 (Cuthbert and Ariza, 2010) (Bach $n = 413$) plus 866 from the Lakh subset (Raffel, 2016) with Tagtraum genre match (Schreiber, 2015) (≥ 15 turns). $\hat{a}$ via sigmoid fit on cum-unique-edges against cum-|interval|/12. The genre hierarchy is perfectly monotone: Rap $\tilde{a} = 0.196 \to$ Electronic $\to$ Pop $\to$ Rock $\to$ Jazz $\to$ Bach 0.409, Mann-Kendall $\tau = +1.000$, $p = 0.003$ (14/15 pairwise Bonferroni-Holm significant); median $R^2 = 0.975$. Genre acts as **material-class**→$\hat{a}$, isomorphic to the timber defect class (§4.13). The pre-registered shuffle control comes out mixed. The expectation was that under note-order shuffle the $\hat{a}$ distribution degenerates to a tight cluster near 0.5 and the genre separation vanishes; the fixed reduction clause was that genre separation surviving the shuffle reduces the finding to a vocabulary effect. What occurred: no 0.5 collapse (the $\hat{a}$ cluster at 0.27–0.34), the monotone genre ordering breaks (Mann-Kendall n.s.; Rap moves from the lowest value into the midfield), and the Jazz level stays separated under shuffle. Under the reduction clause this yields: the monotone ordering is sequence-carried, the

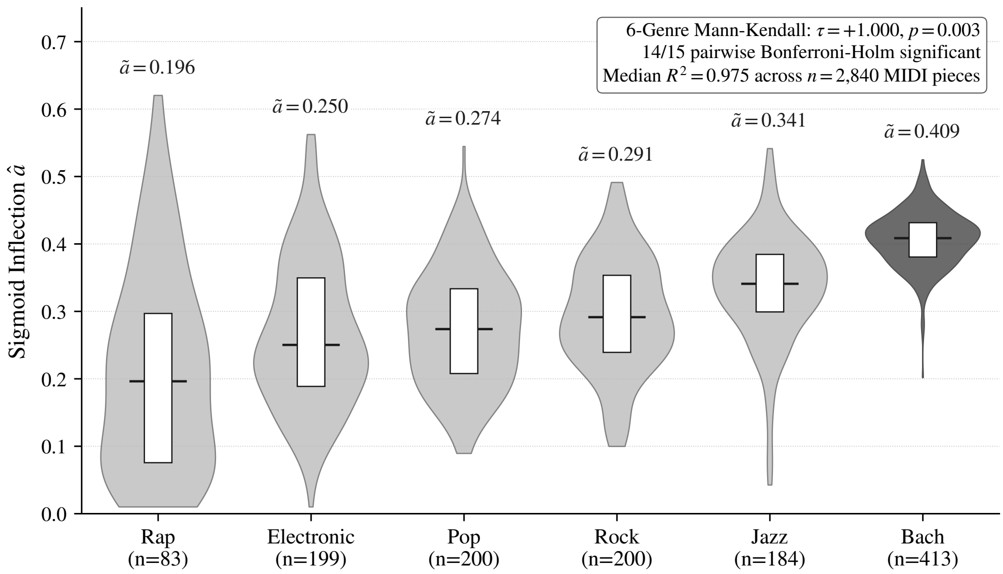


Figure 6: Genre hierarchy across six macro-genres: Rap ($\tilde{a} = 0.196$) → Electronic → Pop → Rock → Jazz → Bach (Classical, $\tilde{a} = 0.409$); Mann-Kendall $\tau = +1.000$, $p = 0.003$. $n = 2{,}840$ MIDI pieces.

Jazz level carries a vocabulary component (the static note multiset). Decisively: Di Marco et al. (2026) (Sci Rep) finds, with an **independent method** (network efficiency rather than sigmoid inflection), the same genre hierarchy, genuine cross-method validation, not borrowed vocabulary.

4.12 Materials-Class Hierarchy

The $\hat{a}$ values order the substrates along an axis of increasing fracture asymmetry, the same sorting logic as DIN 4074-1 (timber S7/S10/S13), DIN EN 10025 (steel yield strength), EN 206 (concrete strength class). The 1D $\hat{a}$-value comparison is illustrative (no-go: $\hat{a} = f(\Pi, \text{form})$, not strictly comparable cross-substrate); the load-bearing statement is the within-substrate correlation, not the absolute value.

Table 6: Materials-class hierarchy with form annotation

Substrate	$\tilde{a}$	Form class / mode	Characterization
Al-6061 (tensile tests, $n = 146$)	**0.084**	Gompertz (form exception)	ductile metal, operating-range ceiling
ENK Chat AI	0.232 ($D = 0.944$)	Logistic / (iv) continuous	conversation (chat-AI turns)
Trecento (14th c., $n = 103$)	0.235	Logistic / (iv) continuous	early tonality development
ENK Conversation (median)	0.268	Logistic / (iv) continuous	LLM substrate (overall median)
Bach (Classical)	0.409	Logistic / (iv) continuous	high baroque
Earthquakes	0.452	Logistic / (i) parallel-instantaneous	foreshock-dominated
Timber (Picea abies, tension)	0.567	Logistic / (i) parallel-instantaneous	fiber anisotropy
Finance SPY DotCom	0.62	Logistic / (i) parallel-instantaneous	speculative asymmetry
Battery (CALB)	0.646	Logistic / (iv) continuous	knee localization (KS) + G7 hysteresis

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Substrate	$\tilde{a}$	Form class / mode	Characterization
Finance SPY GFC	0.97	Weibull / (ii) serial-cascaded	liquidity shock, late inflection
DP1180 (Numisheet)	0.975	Weibull / (ii) serial-cascaded	brittle fracture high-strength steel (KS✓; outcome underpowered, small n)

Form class and mode stem from the companion model comparison (AIC over Logistic/Gompertz/Weibull/Hill; `srds-enk-companion/data/model_comparison/`), in which Logistic dominates 6 of 8 substrates and Al-6061 (Gompertz) and CCP60 (Weibull) are the no-go-predicted exceptions. Each carrier yields along its own path; the load redistributes as soon as one axis reaches its material limit.

4.13 Cross-Substrate Status and Mutual Support

Table 7: Bonferroni status across the ten substrates; political speeches separately as the support bridge to §5 (not counted among the ten)

Status	Substrates
Bonferroni-conformant (γ_M)	Timber, ENK Conversation, Earthquakes, Battery (knee+hysteresis), Music, V-Dem (sigmoid R^2 = 0.983)
Confirmation (accumulation/isomorphism axis)	Multi-Material Multiaxial Fatigue ($\hat{a}$ reads non-proportionality, Chelton in the ENK band; cross-material outcome pooled-damped)
Nominal, caveat named	Lignin (cross-sectional), Colorectal ($\hat{a}$ $\rightarrow$outcome open), Finance (episode-selective)
Support prediction	Political speeches (Bundestag truss null; bridge directionally confirmed, §5)

In addition, there are a pre-registered null control (ProsocialDialog L1) and two falsified positive substrates that map the scope conditions (Sotopia A0, Google COVID Mobility $\sigma\perp\varepsilon$, §8). The substrates do not stand side by side but support one another via shared mechanical axes:

- **Hysteresis / irreversibility (G7):** battery voltage hysteresis ($+0.886$) · multiaxial-fatigue loop · V-Dem recovery (67%, never full): *one* Bauschinger/dissipation mechanics across three substrates.
- **Near-elastic gate:** timber near-elastic ($R^2 = 0.994$) · battery pre-knee: nonlinearity *gates* the yield signal; where there is no plastic regime, no signal.
- **Material-class$\rightarrow\hat{a}$:** timber defect class (S7$\rightarrow$S13) · CRC mutation class (MSI/MSS) · music genre (Rap$\rightarrow$Bach): the same structure$\rightarrow$inflection$\rightarrow$outcome prediction across three substrates.
- $\hat{a}$ **localizes failure, not strength:** earthquake mainshock ($+0.673$) · battery knee (KS): valid localization, weaker strength prediction, same pattern.

The structural statement carries not via the recurrence of a number but via the recurrence of the functional form with substrate-specific asymmetry and the mutual support across these axes: the signal appears where the mechanics predicts it and is absent where it predicts its absence.

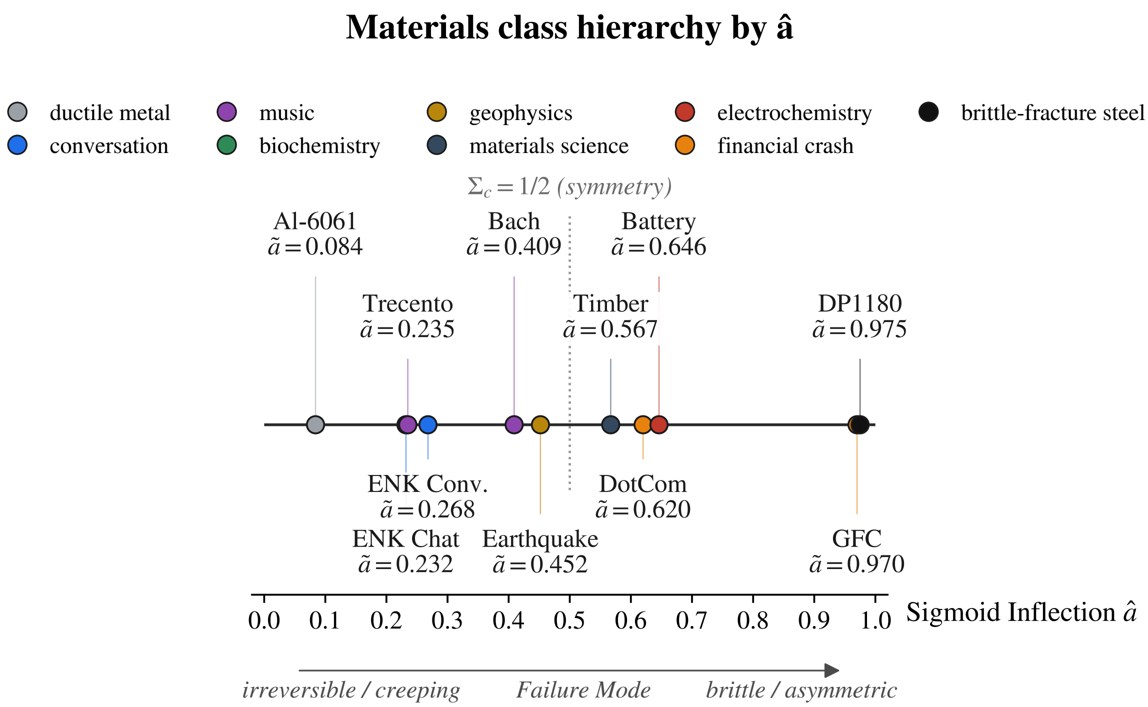

Figure 7: Materials-class hierarchy as a 1D $\hat{a}$-axis: from ductile Al-6061 ($\tilde{a} = 0.084$) to the brittle high-strength end ($\tilde{a} \approx 0.97$). Shown are $\hat{a}$-positions across material classes incl. reference specimens (Al-6061, DP1180, Trecento, financial crashes); V-Dem, CRC, multiaxial fatigue and lignin (no endurance limit, $m < 1$) have no comparable single $\hat{a}$ and are not plotted here. The dotted line marks the mirror-symmetric inflection value $\Sigma_c = 1/2$ (no-go), from which substrate-specific asymmetry is measured; the 1D comparison is illustrative.

5 Structural Scale, Boundary Conditions, and Addressee-Support

The material level from §3 measures the deformation of the individual specimen. The structural level installs this specimen into a configuration with supports, load paths, and internal forces. The boundary-condition typology is the map of these configurations; it decides which of the five measurement axes remains load-bearing and which falls into a zero-force member under the given support. A null signal is then not measurement failure but a prediction of the framing.

Two levels, two operators. The material level (ℓ_2) operates on the stress response of the specimen itself; under multiaxial loading a principal-axis transformation (Mohr, §3) would be the canonical pre-processing, under uniaxial loading it degenerates to the identity. The structural level ($\ell_2 \to \ell_3$) operates on the topology of the support: whether a member is clamped, cantilever, truss, or two-span beam is not a material property but a structural configuration. The two levels are orthogonal: the material principal-axis choice decides which stress components enter the measurement signal; the support topology decides which measurement axes are activated at all. Confusing the two would confuse materials testing with structural engineering.

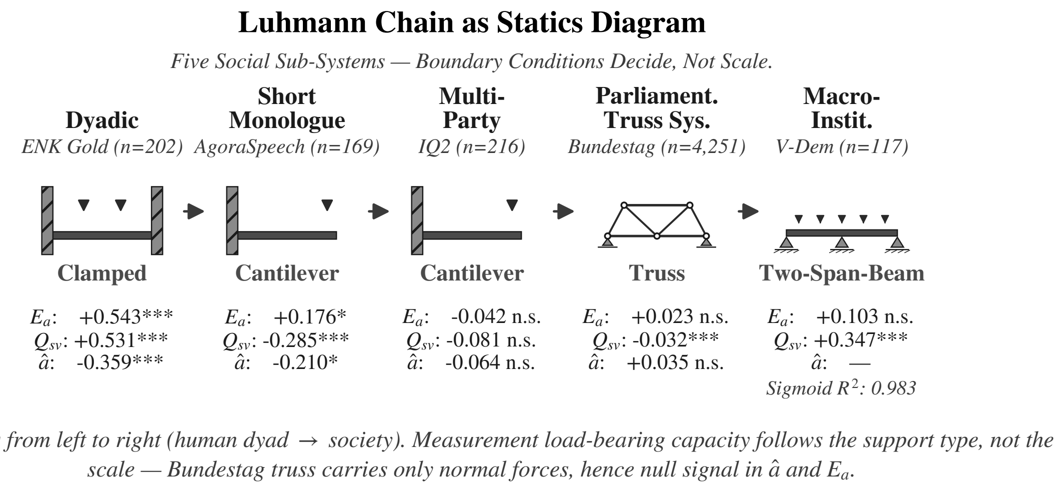

Figure 8: Luhmann chain as statics diagram: five societal sub-systems with their support type (clamped, cantilever, truss, two-span beam). The boundary condition decides, not the scale.

5.1 Five Scales, Five Sub-Systems: the Luhmann Cut

Table 8: Scale-transfer profile across five societal sub-systems

Scale	Corpus	n	Support type	E_a	Q_{sv}	$\hat{a}$	Sigmoid R^2
Dyadic	ENK Gold	202	**Clamped**	$+0.543^{***}$	$+0.531^{***}$	-0.359^{***}	—
Monologue (short)	AgoraSpeech	169	**Cantilever**	$+0.176^{*}$	-0.285^{***}	-0.210^{*}	—
Multi-party debate	IQ2	216	**Cantilever**	-0.042 n.s.	-0.081 n.s.	-0.064 n.s.	—
Parliamentary truss	Bundestag*	4,251	**Truss**	$+0.023$ n.s.	-0.032^{***}	$+0.035$ n.s.	—
Macro-institutional trajectory	V-Dem	117	**Two-Span-Beam**	$+0.103$ n.s.	$+0.347^{***}$	—	**0.983**

All turn-pair statistics are computed cluster-robust; the pooled variants overstate the confidence of the cumulative accumulator family (mean lag-1 $\bar{\rho}_1 = 0.928$; corpus-wide by 42%, Schessl, 2026a). **Bundestag subset* $n = 4{,}251$ *after minimum sampling* $n_{turns} \geq 15$ *(§3.5) from the corpus of* $n = 6{,}874$ *in §4.6.*

The profile does not decay monotonically with scale: it orders by support and load-bearing capacity. The dyadic specimen carries all three quantities, the macro-institutional V-Dem delivers the strongest sigmoid R^2 of the work; the extended Bundestag corpus falls into the structural null range on E_a and $\hat{a}$ (truss, no moment). The five-row structure is motivated by the subsystem decomposition of systems theory (Luhmann, 1984): five carriers, not five arbitrary scales. The support gradient clamped $\to$ cantilever $\to$ truss $\to$ two-span beam is itself a mutual-support axis: the same mechanics (the boundary condition decides which axis bears) across four scales.

5.2 Five Support Types: Structural Mechanics of Conversation

In materials testing, the boundary condition decides which displacement quantities are measurable, which support reactions arise (V vertical, H horizontal, M moment), and which internal forces the system carries (N normal force, Q shear force, M bending moment). The five support types correspond to the standard statics typology (Hibbeler, 2012; Albert et al., 2018; Krätzig et al., 2014).

Table 9: Structural-mechanics signature of the five support types ($V/H/M$ = support reactions; $N/Q/M$ = internal forces)

Support type	Support reactions	Transferred internal forces	Static determinacy	Substrate
Clamped	$V + H + M$ at both ends	N, Q, M	statically indeterminate (3-fold)	**ENK dyadic**
Cantilever	$V + H + M$ at the clamped end, free end	N, Q, M (max. M at the support)	determinate	**AgoraSpeech, IQ2**
Truss	$V + H$ at hinged nodes, no M	only N (member normal forces)	determinate if $2j = m + r$	**Bundestag** (corpus as truss system)
Two-Span-Beam	V at three supports	N, Q, M with mid-support moment	indeterminate to 1st degree (residual stresses, hysteresis)	**V-Dem trajectory**
Pinned	$V + H$, no M	N, Q, no M	determinate with roller	**Political speeches in general**

Static determinacy as a diagnostic axis. Three states can be read off every substrate: *determinate* (support reactions uniquely determined from equilibrium, unique load path: cantilever speeches), *indeterminate* (redundant constraints produce residual stresses that carry hysteresis: doubly-clamped gold-baseline dyad, crystallized conversation, V-Dem recovery asymmetry), *underdetermined* (no unique load path under free movement: diffuse conversation, ProsocialDialog recovery).

Axis-to-internal-force. The split into internal forces $N/Q/M$ is the structural reading of a load response that is uniform at the material level: the micro-imperfection cascade (§2.4) carries every load down to the tensile separation of the smallest bond, and only the support topology decides which measurement axis breaks through. E_a (stiffness) and $\hat{a}$ (yield inflection) break through only where the support transmits a bending moment M, i.e. where bidirectional load redistribution through a counter-support is present; Q_{sv} carries the normal-force path N, which is retained even in a pure truss. Table 8 then reads in advance: where the counter-support is missing (cantilever) or the bending moment vanishes (truss), E_a and $\hat{a}$ fall into the zero-force member while Q_{sv} carries; only the two-sided clamping (clamped) bears all three.

Clamped: dyadic conversation. Both interlocutors deliver full support reactions; all five measurement axes are load-bearing (E_a, Q_{sv}, $\hat{a}$ conformant).

Cantilever: component, not structure. A single speech *taken on its own* is a cantilever: the speaker loads, but an internal counter-support is missing. The axes that measure bidirectional coupling fall into the zero-force member: a yielding without immediate counter-support *within the speech scale*. AgoraSpeech carries a weak residual signal from the paragraph structure; IQ2 falls cleanly into the zero-force member (all three axes n.s.).

Truss: Bundestag as truss system. The corpus over 4,251 speeches is plotted as a truss system: each speech a member, hinged via the agenda as a node. The support reactions carry only $V + H$, no moment; the truss transfers only normal forces. Axes that presuppose a bending moment or hysteresis become zero-force members on the truss. The predicted null on E_a and $\hat{a}$ is structural mechanics, not measurement failure: a truss bears no bending. The effect of the speeches runs via other truss systems (audience, voting behavior, V-Dem indices) and is measurable at the next scale.

Two-Span-Beam: V-Dem. The index trajectory over time has three load-bearing supports (initial, crisis, final constitution), indeterminate to 1st degree $\rightarrow$ residual stresses from path-dependent deformation. Observed: 67% of episodes show recovery within a decade, never identical to the initial state: the hysteresis is the residual-stress signature of the second support. The sigmoid fit reaches $R^2 = 0.983$.

Pinned: political speeches in general. Campaign speeches without a dynamic audience response carry $V + H$, no moment; a methodological intermediate stage between cantilever and truss.

5.3 Addressee-Support: the Measurement Configuration as an API Axis

The five support types are implicit in the SRDS frame. Formalizing them explicitly as a measurement axis makes the configuration visible for applications, analogous to construction-materials tables (S7/S10/S13 for timber per DIN 4074-1; XC1/XC2/XD3/XF4 as exposure classes per EN 206). The SRDS API specifies the addressee classes as an extension and implements them operationally.

Table 10: Five API axes for SRDS measurement configurations

API parameter	Values	Construction analogy	Measurement consequence
`mode`	dyad / multi-speaker / monologue	number of supports	which component topology
`exposure`	XC1 / XC2 / XD3 / XF4	exposure class (EN 206)	which load-case corridor
`tier` (preset)	default / therapy / negotiation / education	sorting class (S7/S10/S13)	which material expectation value
`addressee`	direct / external_modeled / collective_proxy	**support type**	**which support type is realized**
`addressee_E`	default / typical_individual / typical_collective	**E-modulus of the support**	**which standard assumption for an invisible support**

Consequence for the speech section §4.6. With the addressee-support model, a political speech is no longer "predicted null" on the speech-language scale but extended-measurable against a modeled societal support proxy: `mode: monologue`, `addressee: collective_proxy` (audience as collective support: voting behavior, media discourse, V-Dem indices), `addressee_E: typical_collective_<era>`. Under this configuration the Bundestag null becomes a cantilever-style measurement against the societal proxy. The ParlaMint pilot (HU+PL+RS+SI) realizes exactly this configuration: speeches as an $\ell_3 \to \ell_4$ bridge against the V-Dem Polyarchy Index. The force transfer is **directionally confirmed**: the NC2 geometry axis bears $\rho = -0.34$ (BCa-CI$_{95}$ $[-0.48, -0.09]$, $p = 0.006$), the $\hat{a}$-discourse axis is negative across all four clusters (HU -0.40, PL -0.52, RS -0.21, SI -0.40; BCa-CI$_{95}$ $[-0.34, -0.15]$ excludes zero). The permutation $p = 0.207$ is starved of tail resolution at $G = 4$ (4! country permutations) and not load-bearing; the dependable statistic is the BCa-CI plus sign convergence across four independent clusters. Open is the full operationalization (triangulation), not the mechanism.

5.4 Propaganda Mechanics under Missing Restoring Force

The addressee-support configuration carries a mechanical explanation of the propaganda phenomenon. A one-sided loading without direct resistance corresponds to a structure without spring stiffness at the free end: the elastic bedding of structural statics (Winkler, 1867; Pasternak, 1954). The load is absorbed, but no restoring force returns the system; mechanically this implies creep instead of fracture. The same statement follows from the theory of communicative action (Habermas, 1981): without an answer-support the validity-checking counter-speech-act is missing, and validity claims are adopted adaptively instead of examined.

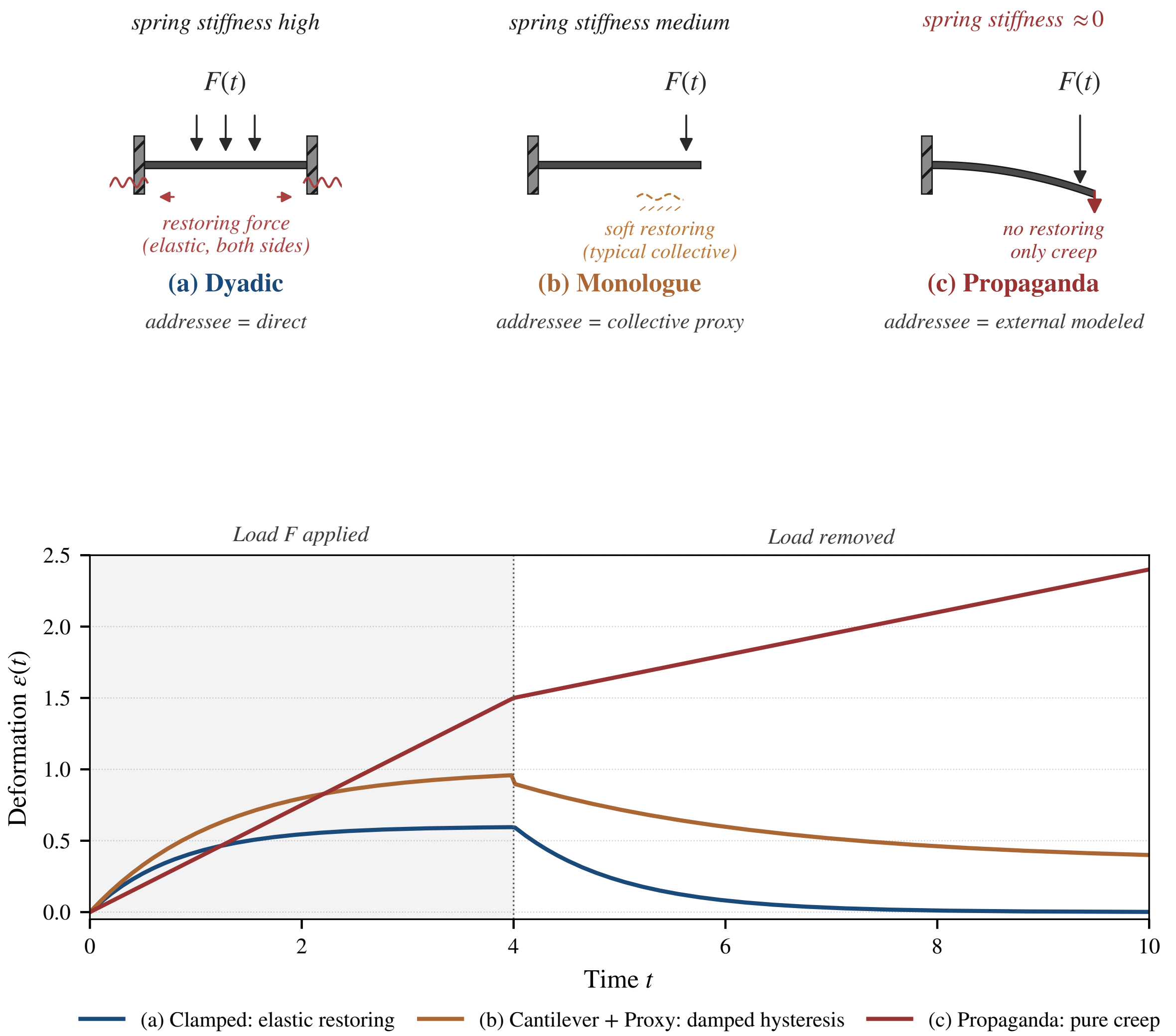

Figure 9: Three spring-stiffness configurations with deformation trajectories: (a) clamped dyadic (elastic restoring), (b) cantilever + collective_proxy (damped hysteresis), (c) cantilever without support (pure creep).

Table 11: Three configurations of spring stiffness at the support

Configuration	Spring stiffness	Response under sustained load	Example
Clamped (dyadic)	high (both sides)	elastic restoring, then plastic deformation with hysteresis	ENK gold-standard dialog
Cantilever with answer-support	medium (collective via `addressee_E`)	damped restoring, slower hysteresis	Bundestag debate with audience
Cantilever without support modeling	**near zero**	**creep without restoring, permanent plastic deformation**	**NS propaganda to an unstructured audience**

Empirical anchor: NS speeches show lower $\hat{a}$ (Cohen's $d = -2.39$, $p < 10^{-4}$, $n = 11$ NS vs. Bundestag $n = 6{,}874$) and stronger frame drift, consistent with the creep model (early yielding without a moderating support). At $n = 11$ the effect size documents direction, not population magnitude; illustrative only. This mechanism is not rhetorical analysis but statics: a structure without spring stiffness at the load point deforms plastically under any one-sided excitation; a democracy with institutional supports carries a moment-load-bearing restoring force that an authoritarian configuration does not provide.

5.5 Empirical Truss Evidence: Institutional Support as Counterforce

The spring-stiffness reading makes a testable prediction: if the institutional truss (horizontal accountability: judiciary/legislature constraining the executive) acts *during* loading as a counterforce that mitigates the deforming load, then a stronger truss should make autocratization slower and costlier without altering the recovery. Across 115 V-Dem episodes in 87 countries (truss measure: judicial-checks index in the year before onset; convergent with the democratic prior history, $\rho = +0.57$), cluster-robust against country clusters:

- **More work to failure (toughness).** Stronger truss $\rightarrow$ higher toughness: $\rho = +0.55$ (judicial checks; cluster-CI $[+0.40, +0.67]$, $p < 10^{-4}$) and $+0.61$ (prior history; $[+0.47, +0.71]$).
- **Slower erosion (damping).** Stronger truss $\rightarrow$ lower steepness k: $\rho = -0.18$ (cluster-CI $[-0.35, -0.001]$, $p = 0.047$) and -0.23 ($[-0.40, -0.04]$, $p = 0.014$).

Both effects are cluster-robust (confidence intervals exclude zero). They are the mechanical signature of the counterforce, not intrinsic material grade: the truss mitigates the *net* load and distributes it across the strut system. Consistently, recovery is *not* truss-modulated. The 67% hysteresis (§5.2) occurs regime-independently (toughness$\leftrightarrow$recovery $\rho = 0.14$ n.s.): the truss acts during loading, not during recovery. On the time-lag axis, the ParlaMint pilot permits a first phase-lag test: the discourse signals lead the polyarchy response by $\approx$ 2–3 years (cross-lagged $\rho = -0.30$ to -0.34 at lag $+3$, $p < 0.03$). This phase lag mirrors the phase shift of transient heat transmission through a storage mass. Four correlated countries, annual resolution: a proof of concept; the cluster-robust measurement across more countries is the companion study.

5.6 Budget State Machine: Architecture without Disclosed Thresholds

At the structural scale, the damage accumulation $B(t)$ from §3.5 is transferred into a graduated state machine:

$$\text{NORMAL} \rightarrow \text{WATCH} \rightarrow \text{WARNING} \rightarrow \text{CRITICAL}.$$

Two configurations operate in parallel: a **production configuration** (ultimate limit state, tuned to a zero false-alarm rate against prosocial reference specimens, strict γ_M) and a **detection configuration** (serviceability limit state, broader sensitivity, nominal γ_M). The structural architecture (four states, two configurations, transition rules) is published. The operational calibration parameters (thresholds, contribution weights, state boundaries) remain withheld: for a deployed stability detector, publishing the exact trigger values would enable targeted evasion (responsible disclosure, §9). No reported finding depends on them: the structural load-bearing capacity is threshold-independent.

Two properties follow from the cumulative structure, not from a configuration: **hysteresis** (evidence accumulates faster than it decays: a loading-unloading asymmetry) and **fatigue memory** (specimens previously in elevated states re-escalate with less additional anomaly; load-bearing capacity does not fully return). Both correspond to established phenomena in materials testing (Palmgren-Miner damage accumulation, Bauschinger effect) and interlock with the G7 hysteresis axis across battery, multiaxial fatigue, and V-Dem (§4.13). The path-dependence is structural, not parametric.

5.7 Application Sketch: ℓ_3-Trajectory Monitoring

Online hate speech often operates in cantilever mode (a single poster loads, with no direct counter-support in the thread) against a modeled collective-support proxy. Message-level detectors (Perspective API, Detoxify) measure the stress peak value, not the cumulative load-bearing capacity of the structure. ℓ_3-trajectory monitoring (thread, account trajectory, subforum corpus) is the structural measurement complement: it is the structural monitor that observes a building over time, rather than a set of single smoke detectors.

Such an application targets trajectory stability, not intent: what a thread shows in cumulative deformation is a material statement about the structure, not about the inner stance of individual participants. Cohen's κ is not the primary validation metric here: what is measured is continuous trajectory deformation, not categorical label agreement, and under the prevailing class imbalance the coefficient collapses paradoxically anyway (Feinstein and Cicchetti, 1990). The load-bearing capacity is established through cross-corpus replication (twelve corpora, §6), not through an inter-rater study. The structural monitor does not replace the smoke detector; it complements it by the second scale, just as a building simultaneously carries acute fire-hazard detectors and long-term settlement sensors (Structural Health Monitoring; Farrar and Worden, 2007; Sohn et al., 2003).

6 Conversation Cross-Corpus and Cross-Method Validation

The material level from §3 measures one conversation on the ENK gold corpus ($n = 202$). That the same deformation profile recurs across twelve external conversation corpora (per-corpus breakdown, including three pre-registered controls, in §6.5) forms the second support of the load-bearing claim. Cross-corpus replication corresponds to multi-cohort validation in clinical epidemiology (Steyerberg, 2009): a method survives when it yields the same measurement signature on disjoint samples under the same structural load.

6.1 Twelve Conversation Corpora (plus Two Cross-Domain Extensions)

Table 12: Cross-corpus validation across 12 corpora

Corpus	n	Mode	Headline statistic	Status
ENK-Gold (anchor)	202	mixed	$\hat{a}$: $\rho = -0.359^{***}$, ridge $\rho = 0.700 \pm 0.076$	PASS
BothBosu v3 (BothBosu, 2024)	1,245	covert	B_w $\rho = +0.317$ $p < 10^{-6}$; permutation 19.2× null median	PASS

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Corpus	n	Mode	Headline statistic	Status
GasConv Gaslighting (Li et al., 2025)	3,998	overt	dir_consistency sign-inverted $p < 10^{-20}$	PASS
Awry (CGA Wikipedia) (Zhang et al., 2018)	3,421	mixed	cross-corpus replication	PASS
LegalCon (Sheshanarayana et al., 2025)	727	covert	**NC2** $\rho = +0.719$**,** $p < 10^{-50}$ (highest singular effect); $\hat{a}$ $\rho = -0.751$	PASS
MentalManip (Wang et al., 2024)	1,294	covert	$\hat{a}$ $\rho = -0.110^{*}$; bond hierarchy PASS	PASS
P4G Persuasion (Wang et al., 2019)	1,016	covert	$\hat{a}$ $\rho = +0.071^{*}$: **sign inversion** (persuasion = later transition)	PARTIAL
SafeDialBench (Cao et al., 2025)	2,037	overt	NCD $d = +6.35$ (highest d value); B_w $d = +3.55$; det_B_t $d = +3.52$; 12/12 sig	PASS
CoSafe (Yu et al., 2024)	1,400	overt	BL-2 Cramér $V = 0.705$; dir_consistency $d = +3.17/-3.80$; ANOVA NCD $H = 62.6^{***}$	PASS
ProsocialDialog	5,000	null result (therapeutic)	$\|\rho\| < 0.10$ all metrics; budget NORMAL in 307/308 specimens	**Null-PASS**
Sotopia agent-agent	3,111	null result (symmetric)	38.7% sigmoid structure, A0 violation	**Null-PASS**
Google COVID Mobility	—	null result (externally imposed)	$R^2 > 0.9$ formal, yet no operator fixed point, no hysteresis	**Null-PASS**
DarkBench / WildJailbreak	—	skip	single-turn = hardness test, not tensile test	construct boundary

In addition, two extended cross-domain corpora from the §5 structure level: AnnoMI (Wu et al., 2022) ($n = 133$ Motivational-Interviewing transcripts, MAS $d = +1.02$ PASS cross-domain, $\hat{a}_{\text{AnnoMI}} = 0.33 \neq \hat{a}_{\text{Manipulation}} \sim 0.52$) and CMV Full (Tan et al., 2016) (Reddit Change-My-View, $n = 11,740$; 8/11 Bonferroni-sig; FID_max partial $\rho = +0.104/d = +0.232$; E_a partial -0.064^{***} survives length control).

Total: $\approx$ 35,300 dialogues across 14 data sources (the 12 corpora of Table 12 plus two cross-domain extensions, AnnoMI and CMV Full): 8 validated, 3 null results, 1 construct boundary, 2 cross-domain extensions. The Sotopia count is the ENK subset with ≥ 8 turns from the cmu-lti/sotopia public release of 7,200 episodes ($n_{\text{Sotopia}} = 3{,}111$); AnnoMI contributes $n = 133$, and the remaining sources total 32,080.

6.2 LegalCon as Showcase: Highest Singular Cross-Corpus Effect

With NC2 $\rho = +0.719$, $p < 10^{-50}$, the LegalCon corpus ($n = 727$, plaintiff-defendant dialogues) carries the strongest singular cross-corpus effect of the entire validation. $\hat{a}$ as strain axis reaches

$\rho = -0.751$. NC2 (Newman conductance on 2-cluster partition) corresponds to grain-boundary density in microstructure analysis (Hall, 1951; Petch, 1953); within LegalCon it measures how the semantic space stiffens under adversarial cross-examination. Here the Hooke coupling runs inverted ($d = -0.881$, brittle-fracture mode), unlike covert manipulation, where it runs in the creep direction. The sign inversion is not a defect but a phase boundary: the same material shows different failure modes under different loading geometry.

6.3 Sign Paradox: Ductile vs. Brittle Fracture

Table 13: Sign inversion covert vs. overt, corresponds to the materials phenomenology of ductile vs. brittle failure modes (Considère, 1885; Bridgman, 1944)

Mechanism	Covert (BothBosu, P4G, MentalManip)	Overt (Awry, GasConv)	Material reading
B_{windowed}	+ (accumulates)	− (discharges)	creeping vs. brittle fracture
FID	+ (frame drifts slowly)	− (topic switches abruptly)	ductile vs. brittle
dir_consistency	− (zigzag, confusion)	+ (confrontation)	confusion vs. consistent confrontation
NC2	+ (expansion)	− (collapse)	stiffening vs. loss of cohesion

The paradox is a classification axis, not methodological ambiguity: the same functional form carries two failure modes (slow creep, abrupt brittle fracture), and the sign inversion tells them apart. This corresponds to the transition from ductile to brittle fracture characteristics in materials testing (Considère criterion for plastic instability; Mohr criterion for brittle fracture). That the same inversion appears between covert and overt manipulation is a measurement on the same substrate, not an analogy.

6.4 Cross-Method Triangulation: Three Methodologically Disjoint Apparatuses

On a D7 phase subset (P3 transition phase), three methodologically independent measurement apparatuses converge on the same phase structure. Method triangulation corresponds to the classical mixed-methods principle (Denzin, 1978; Flick, 1992): a finding holds when it remains replicable across different measurement axes with different formal preconditions.

Table 14: Three disjoint apparatuses on the same substrate

Apparatus	Formal basis	Finding across three phases P1→P3
TF-IDF/SVD norm statistics	linear algebra, singular-value spectrum	NC2-CoV: drop down to −32.5% at SVD-130, robust [80, 150]
Manifold learning TwoNN	Levina-Bickel MLE for intrinsic dimension (Levina and Bickel, 2004), GRIDE family (Denti et al., 2022); LLM-layer adaptation (Viswanathan et al., 2025)	intrinsic dimension monotonically falling
Topological data analysis	Vietoris-Rips filtration, persistence homology (Edelsbrunner and Harer, 2010); zigzag persistence on LLM trajectories (Gardinazzi et al., 2025)	persistence lifetime **1132** → **901** → **824** monotonically falling

The three apparatuses are methodologically disjoint: linear spectra, nonlinear manifold learning, topological filtration. They converge on the same structural statement: phase transitions in the D7 phase subset show a monotonically falling dimension / complexity signature. Cross-method convergence corresponds to the epistemic requirement of measurement robustness against method artifacts (Cronbach and Meehl, 1955, construct validity); no single apparatus proves the finding; the three together, each with its own falsification condition, supply the support.

Concrete drop signs and magnitudes are subset-specific (on a second subset NC2 shows the inverse direction), which supports the no-go statement from §2.5: the asymmetry is substrate-specific, not a universal constant. The triangulation runs on a DUA-restricted ENK subset (participant conversations); the three apparatuses rest on published methods (Table 14), scripts and aggregate results on request.

6.5 One Predicted Null and Two Scope Conditions: Lemma Violations as Construct Support

Three conversation corpora carry no load signal. ProsocialDialog was registered as a negative control *before* measurement (L1 violation, §2.2 Table 2): a genuine predicted null. Sotopia and Google COVID Mobility were pre-registered with directional positive hypotheses (29 March 2026) and falsified; the lemma violations (A0 and L3/$\sigma \perp \varepsilon$, respectively) account for the falsification *post hoc* and map the framework's scope conditions.

Table 15: One pre-registered null control and two falsified positive substrates (scope conditions)

Negative corpus	Violated lemma	Consequence	Finding
ProsocialDialog ($n = 5{,}000$; Kim et al., 2022)	**L1** (Lyapunov half-order) violated $\Rightarrow$ A2 not derivable	$\beta \to 0$, recovery balances damage	all SRDS metrics $\lvert\rho\rvert < 0.10$; budget NORMAL in 307/308 specimens; **LlamaGuard false alarm** $\rho = +0.327^{***}$ on identical data
Sotopia agent-agent (Zhou et al., 2023)	**A0** violated (symmetry instead of asymmetric self-reference)	no path asymmetry possible	38.7% sigmoid structure, fracture characteristic does not occur
Google COVID Mobility	**L3** / $\sigma \perp \varepsilon$ violated	σ and ε both time-driven (circularity)	pre-registered-positive, falsified: raw $r = +0.592$ vanishes after calendar-time control ($R^2 > 0.9$ formal sigmoid, yet no operator fixed point)

Null and positive findings are situated under the same axioms: the same formal apparatus that measures load-bearing capacity on positive substrates *predicts* a null signal on a substrate pre-registered as a negative control (ProsocialDialog). This corresponds to Popper's falsifiability criterion (Popper, 1959) in constructive form. Sotopia and Google COVID Mobility were pre-registered-positive, falsified, and map the scope conditions. Quantified in §8 as a balance: one predicted null plus two scope conditions.

In addition, the DarkBench/WildJailbreak skip: construct boundary, not measurement failure. Single-turn specimens correspond to a hardness test (Brinell, 1900; Smith and Sandland, 1922), not a tensile test (DIN EN 408). The ENK $\hat{a}$ methodology requires a loading history of at least 15 turns (minimum sampling from §3.5); single-turn lies below this minimum load duration. The skip is construct-consistent: a hardness test cannot be read as a tensile test.

7 Cognition Bridge and External Frame Convergence

7.1 Cognition Bridge: G_f/G_c and the Sigmoid Mélange

The sigmoid inflection $\hat{a}$ bridges to a cognitive carrier class; this is a theoretical extension, not yet empirically validated. The Cattell-Horn-Carroll model in intelligence psychology (Cattell, 1963; Horn and Cattell, 1967; McGrew, 2009) distinguishes fluid intelligence G_f (adaptive-elastic, new problems) from crystallized intelligence G_c (accumulated-brittle, experience-based). The life-span trajectory is empirically documented: G_f falls concavely from early adulthood, G_c rises convexly into late adulthood (Salthouse, 2004).

Material reading: G_f corresponds to elastic deformation (reversible, adaptive load redistribution), G_c to plastic deformation (accumulated, irreversible structure formation). The intersection of the two trajectories, the *mélange*, has sigmoid form: a transition from G_f-dominated to G_c-dominated with inflection point in middle adulthood. A sigmoid fits this mélange form ($R^2 = 0.853$). Unlike the empirical fits for V-Dem ($R^2 = 0.983$) and earthquakes ($R^2 = 0.965$), however, this fit measures agreement with the *theoretical* CHC mélange, not with longitudinal data. Validating this against longitudinal data (HRS/MIDUS) remains follow-up work. Here $\hat{a}$ marks the yield point at which accumulated experience (G_c, plastic) compensates for the slowing rate of adaptation (G_f, elastic): wisdom compensation in life-span psychology (Baltes and Staudinger, 2000; Baltes and Smith, 2008).

The bridge makes a cognitive carrier class visible as a *hypothesis*, alongside the empirically measured substrates, not as an eleventh validated substrate. Operationalizing the trajectory as a sigmoid is a hypothesis with a defined falsification condition (HRS/MIDUS longitudinal), not an analogy.

7.1.1 Chen et al. 2026: Long-CoT as Molecular Structure

A second bridge operationalizes the molecular structure of thought directly in the cs.CL substrate. Chen et al. (2026) (*The Molecular Structure of Thought*, ByteDance Seed / HIT / Peking University, arXiv:2601.06002) model long-chain-of-thought trajectories as molecule-like structures with three bond types:

Bond class (Chen 2026)	Materials-science reference	ENK analogue
Deep reasoning	covalent bond (~ 350 kJ/mol)	$1 -$ sem velocity (long-range carrier structure)
Self-reflection	hydrogen bond (~ 12 kJ/mol)	h_{bond} decay ($\rho = +0.398^{***}$, LOPO 6/6)
Self-exploration	Van der Waals (~ 3 kJ/mol)	mean-cos far-field ($\rho = +0.133$, n.s., $p = 0.062$; theory-consistent weak for the weakest bond class)

Chen's Effective Semantic Isomers carry the statement that *only bond configurations with fast entropy convergence* support stable long-CoT learning, while *structural competition* destabilizes training. This is structurally homomorphic to the damage signature measured here as B_w accumulation (§3.5). This corresponds to the Anfinsen hypothesis of protein folding (Anfinsen, 1973): just as the tertiary structure is encoded in the amino-acid sequence, the stable CoT trajectory is encoded in the bond configuration of the reasoning steps. The four-level hierarchy (sequence, secondary structure, tertiary folding, multimer; (Berg et al., 2019)) corresponds in the language substrate to the staggered levels from token sequence through local patterns and thematic clusters to multi-conversation behavior, paralleling the scale hierarchy ℓ_0 to ℓ_4 from §2.3.

7.2 External Frame Convergence: The Gap That SRDS Fills

Several independent groups arrive, under different disciplinary names, at structurally identical arguments. **These works do not *confirm* SRDS**: they show that the same class form arises independently across disjoint fields. The list is not the load-bearing pillar of this work; it is evidence that the gap a cross-substrate frame closes is real. These convergence anchors satisfy the epistemic requirement that a theory be robust against observational artifacts (Hacking, 1983).

Table 16: Convergent independent work across disjoint fields

Anchor	Substrate	Statement	Reference to SRDS
Olmeda et al., 2026 (Nature Physics)	epigenome	self-similarity power-law $5/2$, renormalization-group schema	RG operator $M = N \circ R \circ C$ (§2.3) on a biological carrier
Simon et al., 2026 (*learning mechanics*, arXiv:2604.21691)	deep learning	discipline-naming in direct parallel to the frame presented here	seven desiderata, including *humble*, meets the construct-boundary concept (§8)
Zhdanov et al., 2025 (AdS-GNN)	conformal-equivariant geometry	operator composition on a mathematical carrier	functor invariance, not numerical fidelity (§2.5)
Schmidt et al., 2026 (Topological Deep Learning)	topological vs. combinatorial generalization	topology as measurement substrate	functor $\Psi : \mathrm{SRDS} \to$ aggregate class as functorial realization
Yang et al., 2026	mesh-agnostic operator composition	operator logic independent of the discretization mesh	renormalization operator M carries under coarse-graining (§2.3)
Bakeer, 2026 (computational mechanics, arXiv:2605.28601)	spatial parameter identifiability	local information operators; two-span support coupling, beam kernels, "visible/invisible" directions	same statics vocabulary on a *spatial* rather than temporal carrier; published later and addressing a different problem: *existence, not confirmation*
Chen et al., 2026 (ByteDance Seed, arXiv:2601.06002)	cs.CL long-CoT	bond topology covalent / H-bond / vdW (§7.1)	mélange operator realized directly in the language substrate

The convergence rests not on count but on structural similarity across disjoint substrates. Olmeda (biological renormalization) and Simon (*learning-mechanics* naming) arrive via disjoint paths at the same form: renormalization operator as central measurement axis, construct-boundary concept as methodological stance. Three mathematical works (AdS-GNN, Topological Deep Learning, mesh-agnostic composition) realize the same functor on disjoint formal carriers; one cs.CL paper (Chen 2026) operationalizes the molecular structure of thought directly in the language substrate. The computational-mechanics anchor (Bakeer, 2026) applies local information operators to the *spatial* identifiability of distributed parameters: the same statics vocabulary (two-span support coupling, beam kernels, visible vs. locally invisible directions) on a disjoint, spatial carrier; later in time and on a different problem, hence convergence, not confirmation. The class SRDS is realized consistently in structure across three mathematical languages: functor-invariant, not number-faithful (§2.5).

Closest to this approach is Buehler's materiomics line: hierarchical structural isomorphism across materials, proteins, music, and language, most recently a self-organized-critical graph-reasoning state with a dimensionless *Critical Discovery Parameter* (Buehler, 2025). The overlap is real and touches two substrates; so is the delta. That line reads *structural* similarity across a single substrate family and *seeks* a critical parameter, whereas SRDS fixes a *dynamical* damage class (load, accumulation, yield inflection $\hat{a}$, irreversibility) across ten physically disjoint substrates: with a synthetic null and pre-registered scope limits (§8), a computer-assisted no-go theorem with Lean-4 formalization (§2.5), and retained failures.

Where that line seeks such a critical parameter, the no-go theorem *excludes* a universal constant.

Beyond structural form convergence, the *methodology* of this cross-substrate move has its own name in philosophy of science: **patchwork concepts** (Wilson, 2006; Haueis, 2024). A concept that fragments into "patches" across scales and substrates (each with its own measurement technique, united by a *general reasoning strategy* rather than an identical core mechanism) is on this account not a mere analogy but a regular conceptual form. This covers exactly what SRDS does at the measurement level: the technique changes per substrate (tensile test, V-Dem fit, embedding geometry, capacity fade), the measured property (inflection↔outcome) remains. Haueis's third norm, *to check whether the property actually applies in each patch* (Haueis, 2024, pp. 749–750), coincides with the PARTIAL/FAIL discipline practiced here: the substrates without signal (ProsocialDialog, Sotopia, Mobility) *support* the conceptual form rather than weakening it. Independently, Rorot (2025) investigates the scale-free concept of *communication* across disciplines (corpus analysis over ∼1.1 M papers) and defends the realist reading against the "mere metaphor" objection: there is "*no clear, arbitrary line*" between literal and metaphorical use, and the realist reading has "*firmer grounds than any anti-realist readings*" (pp. 265–266). This licenses the cross-substrate measurement step, not a shared mechanism: the *deeper* unification remains the RG folding onto atomic Mode-I tension (§2.3, §9.1).

8 Falsification Anchors, State of Measurement, and Limits

The burden of proof is reversed (§1.1b): the mechanics is not defended case by case; the skeptic would have to exhibit a finding that *violates it with adequate load and specimen count*. This section states the falsification tests (§8.1), the three falsification layers (§8.2), the fit↔counterpart systematics (§8.3), the comparison against message-level detectors (§8.4), the limits (§8.5), and closes with the structural limit (§8.6).

The demarcation. A **falsification** is a finding that **violates a physical prediction with adequate load and specimen count** (sufficient statistical power to reach a yield signal): a severed yield-point/strength coupling, a bending truss, a self-reversing accumulation. **Not** a falsification is (a) an underpowered null (small sample, wide confidence interval), (b) a *moot* test (on forced sequences a path surrogate is physically meaningless, §1.1b), or (c) a near-elastic specimen without a plastic regime (no yield → no yield signal). An underpowered null is also **not** a "partial confirmation". Each is named for what it is.

8.1 The Falsification Lattice

Each row names a physical rule, the finding that would *overturn* it, and the observation. None is overturned.

Table 17: Falsification lattice (rule → falsifier → observed)

#	Physical rule	What would falsify it	Observed
G1	forced → material law holds	timber $\hat{a}$ says nothing about strength	$r(\hat{a}, \sigma_{\text{max}}) = -0.83$, $p_{\text{perm}} < 5 \times 10^{-4}$
G2	free → the load *path* carries the signal	ENK $\hat{a}$↔mal survives path destruction	$z = -4.86$ (stands out of the surrogate cloud)
G3	a truss bears no bending moment	strong $\hat{a}/E_a$ signal at the Bundestag truss	$E_a + 0.023$ n.s., $\hat{a} + 0.035$ n.s. (null member)
G4	no universal constant (no-go)	*one* universal $\hat{a}$ value across all substrates	KS $\hat{a}_{\text{emp}} \neq 0.50$ in 10/10 distinct ($p < 0.01$; 7/10 at $p < 10^{-3}$)
G5	$\sigma = E\varepsilon$, axes coupled	$\hat{a}$ and amplitude perfectly orthogonal	$\rho(\hat{a}, \text{std}) = +0.807$, $z = +11.2$ (path-generated)

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(continued)

#	Physical rule	What would falsify it	Observed
G6	"the sigmoid is trivial"	sigmoid wins also on random monotone curves	AICc: power law 9/10, sigmoid-4p never dominant
G7	damage accumulates irreversibly	budget decays faster than it accumulates	hysteresis: V-Dem 67% · battery $V_C - V_D$ $+0.886$ · multiaxial
G8	substrate-specific → $\hat{a} \leftrightarrow$ outcome not equal everywhere	permutation $r(\hat{a}, Y)$ identical everywhere	substrate-specific: timber, chat-AI, FKM, finance carry; near-elastic null (no yield) + DP1180 (underpowered) do not

8.2 Three Falsification Layers

Layer 1: synthetic null specimens. 1,000 specimens per substrate (AR(1)/RW/WN) analyzed with the same pipeline; on all five substrates tested against the synthetic null (Table 18) the Kolmogorov-Smirnov test separates at $p < 10^{-3}$; the sectoral finance sub-structures bear the load more weakly the more loosely they couple to their index.

Table 18: KS-test against synthetic null, five substrates (all with $p < 10^{-3}$)

Substrate	KS-D	p	Component type
Timber DIN EN 408 ($n = 230$)	0.801	$< 10^{-3}$	timber tensile test
Al-6061 ($n = 146$)	**1.000**	$< 10^{-3}$	ductile metal ($\tilde{a} = 0.084$)
DP1180 Numisheet	1.000	$< 10^{-3}$	high-strength steel
ENK Chat AI ($n = 202$)	0.944	$< 10^{-3}$	LLM conversation
Battery CALB ($n = 26$)	0.733	$< 10^{-13}$	Li-ion degradation

Layer 1b: model comparison against the saturation class family (K1). The sharper objection holds that a normalized monotone cumulative is almost always S-shaped, so the comparison against AR(1)/RW/WN is too weak; the *correct* null is some smooth saturation function. The model comparison across the closed class family (Logistic, Gompertz, Weibull CDF, Hill, AICc) settles this: on random monotone curves the 2-parameter power law wins in 9 of 10 domains and the 4-parameter sigmoid is **never dominant**; hence the sigmoid is not privileged against the correct null. On the empirical substrates Logistic is the best fit in 6 of 8, with substrate-specific form as a No-Go-predicted exception (Al-6061 → Gompertz, $\rho(\hat{a}, \sigma_{\max}) = -0.703$ on 146 validly fitted of 154 tensile specimens; CCP60 → Weibull, $\rho = +0.867$). The inflection–outcome relation is robust across all four model families (cross-model $\rho = 0.87$–0.99). The corresponding pre-registration carried three primary form hypotheses, **none confirmed** (no universal Logistic dominance); the secondary hypotheses (inflection–outcome substance, cross-model robustness) are confirmed.

Layer 2: predicted null and empirically found scope conditions. ProsocialDialog was registered as a negative control *before* measurement (L1, $\beta \to 0$, §6.5) and carries the predicted null: no false alarm despite therapeutic intensity ($\rho = -0.008$). Sotopia and Google COVID Mobility, by contrast, were pre-registered with *directional positive hypotheses* (29 March 2026) and falsified; their failure empirically maps the framework's scope conditions: Sotopia violates A0 (agent-agent symmetry instead of asymmetric self-reference); Google COVID Mobility violates the $\sigma \perp \varepsilon$ premise (mobility and cumulative stringency are both time-driven, a circular operationalization). Ductile metals additionally mark the operating-range ceiling. A *pre-registered* null is the sharpest falsification condition; a *falsified positive prediction* that reveals a scope limit is the second-sharpest; neither can be produced by a curve-fit artifact.

The axiom reading of the two limits is a *post-hoc* explanation of the falsification, not a pre-registered null.

Layer 3: sealed test batch. Pre-registration (18 March 2026), six primary hypotheses under Holm-Bonferroni. On the strictly separated primary batch never used for calibration (batch A, $n = 36$) all six predictions lie in the predicted direction; under the conservative Holm correction **one** clears the single-metric threshold (H3, $r = +0.140$). **The decision rule $\geq$4/6 was not reached**: reported as such, not as "partial confirmation". Alongside the six single metrics (primary hypotheses), the accumulator was pre-registered as a *meta-hypothesis*: this meta-hypothesis (number of cluster-robust correlations against chance) holds on both batches (batch A $4.5\times$, batch B $6.3\times$ against the pre-specified $2\times$ threshold). The inflection point $\hat{a}$ carries $\rho = -0.566$ on the second sealed batch ($n = 46$ valid of 50), a *non*-pre-registered additional finding, named as such (not "pre-reg confirmed"). That all six point in the predicted direction under limited test load and specimen count is the signature of a real effect at small sample size; a null effect would scatter the signs.

8.3 Fit$\leftrightarrow$Counterpart: the Nulls Are the Proof

Every positive fit has a *counterpart*: a predicted null where the signal must mechanically be absent. A frame that found signal everywhere could not produce these counterparts.

Table 19: Fit$\leftrightarrow$counterpart across substrates

Substrate	Positive fit	Counterpart (predicted null/boundary)	Demarcated
Finance	crash sigmoid $R^2 = 0.97$	calm = random walk $R^2 = 0.48$ ($p = 0.0001$)	"every system saturates" refuted
ENK	real path $z = -4.86$	surrogate cloud (path destroyed $\rightarrow$ generic)	path mechanism vs. generic accumulation
Timber (forced)	$\hat{a} \leftrightarrow \sigma_{\text{max}} = -0.83$	near-elastic cohort, $R^2 = 0.994$ (no yield)	plastic regime *gates* the yield signal
Model form	sigmoid on real $\sigma(\varepsilon)$	power law 9/10 on random curves	sigmoid not a fit artifact (§8.2)
Structure	V-Dem two-span $R^2 = 0.983$	Bundestag truss $\hat{a}/E_a$ null ($+0.035$ n.s.)	support topology, not scale (§5)
Negative control	scam $19.2\times$ (fires)	ProsocialDialog $\rho = -0.008$ (no false alarm)	dose by load, not everywhere
Pre-reg null / scope limit	(load corpora fire)	Prosocial L1 (pre-reg null, confirmed); Sotopia A0 / Mobility $\sigma \perp \varepsilon$ (positive pre-reg falsified)	one pre-reg null + two empirical limits

The pattern is invariant: the signal appears *where the mechanics predicts it* and is absent *where the mechanics predicts its absence*. The latter is the sharper statement: a curve-fit artifact produces no predicted null.

8.4 ENK against Message-Level Detectors

Head-to-head on subtle manipulations ($n = 8$): SRDS 5/8, LlamaGuard 2/8, one tie. On ProsocialDialog ($n = 5,000$, of which 308 with $n_{\text{turns}} \geq 15$) SRDS monitoring delivers $\rho = -0.008$ (zero false alarms; budget NORMAL in 307/308 specimens), while LlamaGuard shows on identical data $\rho = +0.327^{***}$: systematic false alarms at therapeutic intensity (Perspective API carries a comparable profile). This is not a detector competition but a pairing of complementary measurement classes: a fire alarm measures acute peaks, a structural monitor measures cumulative load-bearing capacity (Tresca peak value vs. fatigue analysis, Wöhler, 1870). Both are needed.

8.5 Limits

Eight limits, each named precisely and placed accordingly. The numbers stand unchanged; the placement only sorts each point as a sufficiently powered falsification, an underpowered measurement, or a construct property.

Table 20: Eight limits, placed

Limit	State	Placement
Bonferroni balance 6/10 conformant	six PASS, Fatigue accumulation-confirmation, three nominal, one predicted null + two scope limits	measurement balance (no re-count)
Annotator dominance 69%	LOPO macro $\rho = 0.339$ vs. length reference 0.174	underpowered measurement, extensible (more annotators)
Calibration thresholds withheld	architecture + axioms + Lean proofs open	disclosure choice (responsible disclosure, §9); structural capacity threshold-independent
Cross-temporal encoder coupling	addressable via `addressee_E` + epoch-specific encoders	open measurement task (ParlaMint shows multilingual scaling)
Encoder-specific signs	different encoders = different asymmetry measures	construct-internal prediction (no-go, §2.5), not a limit
Cross-group replication missing	four reproducibility layers documented (protocol/apparatus/corpus/method, §6.4)	underpowered measurement, cross-group = 5th layer (outreach)
$\hat{a}$ interpretation substrate-specific	no universal $\hat{a}$ $\rightarrow$outcome mapping	construct-internal prediction (universality classes, Wilson, 1979)
Sealed test batch: $\geq$4/6 (batch A) not reached	accumulator meta holds ($4.5\times$/$6.3\times$); $\hat{a}$ on second batch $\rho = -0.566$	underpowered measurement (§8.2), not a falsification; extensible

In sum: predominantly state of measurement (extensible), plus two construct-internal predictions from the no-go theorem itself and one disclosure choice. No structural limit addressable by additional data, except the no-go (§8.6).

8.6 What Remains: The Structural Limit

After review of the limits, **one** structurally indispensable limit remains: the no-go theorem (§2.5). It rules out that Σ_c carries as a universal constant across all admissible kernels. This limit is not addressable by data or methods; it is *part of the theory itself* and is published in the companion. Structurally this corresponds to Gödel incompleteness (Gödel, 1931) in the sense in which A0 (§2.2) takes up self-reference: the frame formally specifies its own universality limit and makes it testable. The measurement grammar describes what the load does to the form, not which load a system *should* carry.

9 What Remains When Something Is Missing

9.1 Summary of Findings

This section draws the threads together. The framework comprises five axioms A0–A4, four regularity conditions L1–L4, three core theorems T1–T3, and one no-go theorem (computer-assisted proof: $\dot{Q}(1/2) \in [1.8412, 1.8963]$). The proof is reproducible with `make verify`; the Lean-4 code is public and the certificate is load-bearing per `#print axioms`. The Lean axiom conservatively encodes $[1.75, 1.99] \supsetneq [1.8412, 1.8963]$, so zero is excluded. The measurement reduces to three steps: accumulate a cumulative strain coordinate, fit a sigmoid, and read off the inflection point $\hat{a}$. Synthetic reference specimens (1,000 per substrate, AR(1)/RW/WN) show that the functional form is not an artifact of fit freedom: on all five substrates tested against the synthetic null (Table 18) the KS test reaches $p < 10^{-3}$, and the model comparison against the saturation class family (Gompertz/Weibull/Hill) does not privilege the sigmoid on random curves (§8.2).

Six of ten substrates meet Bonferroni conformity under the conservative γ_M (timber, ENK conversation, earthquakes, battery under caveat, music, V-Dem sigmoid load-bearing capacity). Multi-Material Multiaxial Fatigue carries as a confirmation on the accumulation axis ($\hat{a}$ reads the load-path non-proportionality, cross-material outcome pooled-damped). Three show nominal signal with named caveats. The scope conditions are mapped by a pre-registered null control (ProsocialDialog L1, confirmed) and two falsified positive predictions (Sotopia A0, Google COVID Mobility $\sigma \perp \varepsilon$).

The main claim rests on empirical evidence. Timber opens with Pearson $r = -0.83$ over $n = 230$ specimens per DIN EN 408:2012-10; the same sigmoid class family recurs in ten substrates, with deviations in form class and inflection location that are themselves measurement values. The spread of inflection points is not deviation from an ideal but a measurement of the substrate's actual asymmetry: timber $\tilde{a} = 0.567$ (fiber anisotropy), battery 0.646 (CALB, KS-validated), DotCom 0.62, GFC 0.97 (liquidity shock), lignin (inflection at the origin, no endurance limit), earthquakes 0.452, Bach 0.409. This scatter is consistent with the no-go theorem, which formally rules out a universal inflection constant: what is universally load-bearing is the functional form, not the numeric value. Mode-I universality anchors this mechanistically: bending produces compression above and tension below, yet the fracture at the crack tip is always atomic Mode-I tension; the mechanism does not rest on a single dataset, and each substrate carries its own falsification test.

9.2 Open Replication Tasks

Independent replication outside this research group remains open; two programs structure these tests.

Program A: Causal intervention. Separation of correlation and mechanism. The crystallization signature ($\chi^2 = 18.813$, $p = 0.0001$, Cramér's $V = 0.310$) and the meta-Fisher across eight corpora ($\chi^2 = 68.96$, $p < 10^{-6}$) deliver structure; an intervention study decides whether they *carry* the load or only resonate: the RCT path of clinical medicine (Hill, 1965, Bradford-Hill criteria).

Program B: Multi-agent symmetry. Two discrete points are established: Sotopia (fully symmetric, 38.7% sigmoid structure, A0 null confirmed) and the ENK gold corpus (asymmetric human-machine dyads, $\rho = -0.359$, A0 active). What remains open is the continuous intermediate stage: slight asymmetries as a parametric gradient, A0 violation as a sweep rather than a single test.

9.3 Data and Code Availability

Table 21: Reproduction anchors (companion paths relative to `srds-enk-companion/`)

Path	Contents
`lean4/PositivityProofs/`	no-go theorem computer-assisted proof (Arb certificate $[1.8412, 1.8963]$, certificate load-bearing per `#print axioms`)
`genealogy_of_frames.md`	dated precursor frame classes (priority anchor)
`data/earthquake,battery,crc,holz_master,lignin,musik_companion/`	cross-domain raw data and pipeline scripts
`data/model_comparison/`, `analysis/`	model comparison (K1, AIC over Logistic/Gompertz/Weibull/Hill) + KS null + permutation inference
`REPRODUCIBILITY_MANIFEST.md`	per table/figure: script path, raw data, external URLs
(ENK subset, DUA, on request)	cross-method triangulation NC2/TwoNN/Vietoris-Rips; methods published + aggregate finding in §6.4
Zenodo 10.5281/zenodo.18340365	timber DIN-52188 data (Schessl 2025)
Zenodo 10.5281/zenodo.5142664	MetaMIDI music source (Ens and Pasquier, 2021)

External substrates rely exclusively on publicly available raw data (USGS-FDSN, TCGA Pan-Cancer Atlas, Yahoo Finance, V-Dem ERT-v14, ParlaMint, Materials Cloud, BatteryLife). No data on identifiable persons outside the documented corpus; conversation data in the gold corpus were collected with consent and are pseudonymized. The central cross-substrate claim is independently reproducible via the twelve external conversation corpora ($\approx$ 35,300 conversations from 14 sources, §6.1) without gold-corpus access.

9.4 Genealogy

The frame class is publicly documented with dates, a priority anchor, not an evidential support:

Document	Date	Repository
Master thesis (timber data anchor)	May 2025	Zenodo 10.5281/zenodo.18340365
Emergent Narrative Control	21 Aug 2025	Zenodo 10.5281/zenodo.17556213
Life as Organized Dissipation (bridge to A1)	26 Oct 2025	Zenodo 10.5281/zenodo.17445178
The Self-Reference Axiom (precursor to A0)	28 Oct 2025	Zenodo 10.5281/zenodo.17509367
ENK: Minimal-Disclosure Monitoring Interface	3 Feb 2026	Zenodo 10.5281/zenodo.18475921
Autocorrelation Blind-Spot	15 Apr 2026	arXiv:2604.14414

These precursor works are consolidated here into the class SRDS; an extended genealogy is documented in the companion as a public dating anchor (`genealogy_of_frames.md`).

9.5 Disclosure

Code availability. Lean-4 proofs and metric code are public (`github.com/FerdinandSchessl/srds-enk-companion`); the master-thesis timber raw data are archived on Zenodo (10.5281/zenodo.18340365).

Data availability. The ENK gold corpus is available on request under a data-use agreement (participant privacy compliance); cross-substrate raw data and reproduction scripts are documented in the companion (Table 21).

Responsible disclosure. The structural architecture of the state machine (four states, two configurations, transition rules), the axioms A0–A4, the theorems T1–T3, and the no-go theorem are fully published. Only the operational calibration thresholds remain withheld: for a deployed stability detector, publishing the exact trigger values would enable targeted evasion. No reported finding depends on them: the structural load-bearing capacity is threshold-independent; the withheld values are bound by a SHA-256 commitment in the companion (`COMMITMENT.md`), making their integrity and dating verifiable without disclosing them. The regime-bound design is not merely a disclosure choice but theoretically forced: a second no-go theorem in the formal companion framework (Schessl, 2026b) shows that under interventional monitoring (when the published verdict feeds back into the subsequent dynamics) no universal static classifier exists; the thresholds are calibration parameters of a regime, not universal constants, and only the state-machine architecture is part of the theory.

Use of generative AI. Generative AI (Claude Opus) was used both in the analysis (gold-annotation reference protocol, §3.3) and in drafting this manuscript; all content, numbers, references, and conclusions were manually checked by the author, verified against the raw data, and are fully the author's responsibility. AI is not listed as an author.

Conflict of interest. The author declares no conflict of interest.

9.6 Why Physics Does Not Stop at Self-Referential Systems

A common reflex treats language as if mechanics made an exception for it: turns are sampled, permuted, and counted as independent observations. That is the category error. A conversation under load is *one* specimen with an ordered, continuous load history, not a bag of interchangeable tokens. Cutting a loaf into thirty slices does not yield thirty loaves but thirty cuts through *one* piece; the lag-1 autocorrelation ($\bar{\rho}_1 = 0.928$, §3.3) *is* the loaf as a continuum, not a nuisance term to be removed. That this treatment is physically correct is no special pleading of this work: geophysics has done the same for half a century.

Hasselmann (1976) reads climate as *integrated* weather: there the temporal autocorrelation (red noise) is the signature of accumulation, a measured value, not an artifact. The same structure carries the ENK accumulator (Palmgren-Miner damage, §5; Schessl, 2026a). What does *not* stop at self-referential systems is continuum mechanics: load, accumulation, yielding, fracture, RG-folded from the atomic Mode-I level up to society (§2.3, §9.1). What stops is only the *independence assumption* of inferential statistics, and for the loaf it never held. The substrate is language; the laws are the same.

9.7 Closing Remark

Simon et al. (2026) has concurrently proposed a programmatically related discipline under the name *learning mechanics*; its construct-boundary concept (*humble*: solid in what is described, explicit in what cannot be described) meets the one used here (§8.6). The frame is not ad hoc but structurally convergent; independent groups arrive at the same form on disjoint carriers: biological renormalization (Olmeda et al., 2026, Nature Physics), *learning mechanics* (Simon et al., 2026), three mathematical realizations of the same functor (AdS-GNN, Topological Deep Learning, mesh-agnostic composition), one cs.CL paper (Chen et al., 2026), and one computational-mechanics paper (Bakeer, 2026); §7.2. The work presented here operationalizes the class on language and tests it on ten substrates across eight research fields.

The no-go theorem (§8.6) formally bounds the universality limit and makes it testable. What must redistribute, deforms; what can no longer redistribute, breaks. What remains is the material.

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